

Motif discovery

String-based approaches

Jacques van Helden

<https://orcid.org/0000-0002-8799-8584>

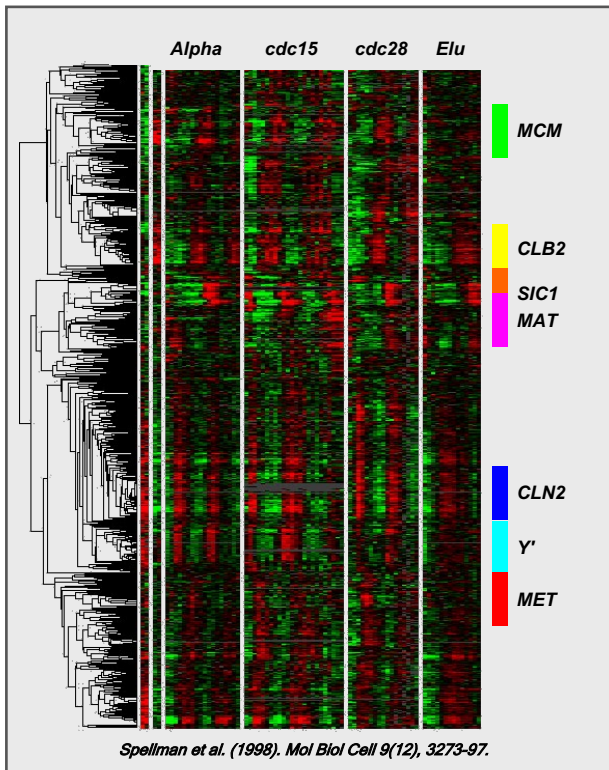
Aix-Marseille Université, France

Theory and Approaches of Genome Complexity (TAGC)

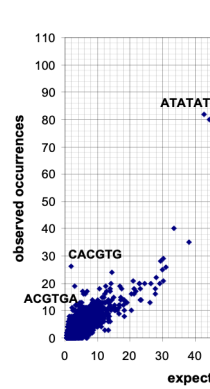
Institut Français de Bioinformatique (IFB)

<http://www.france-bioinformatique.fr>

Motif discovery in promoters of co-expressed genes



1. Count k-mer of 2. Over-representation statistics



seq	identifier	exp_freq	occ	exp_occ	occ_P	occ_E	occ_sig	rank
ccacag	ccacag ctgtgg	0.0002365577659	18	1.97	4.8e-12	1.0e-08	8.00	1
cacgtg	cacgtg cacgtg	0.0001569968432	14	1.31	1.4e-10	3.0e-07	6.52	2
actgtg	actgtg cacagt	0.0003762020409	20	3.13	1.7e-10	3.6e-07	6.45	3
aactgt	aactgt acagtt	0.0006168655788	21	5.14	1.2e-07	2.6e-04	3.59	4
tggtg	tggtg tggtg	0.0002897084237	14	2.41	2.8e-07	5.7e-04	3.24	5
acgtt	acgtt acgtt	0.0003355962588	15	2.80	2.8e-07	5.8e-04	3.24	6
					1.7e-05	3.6e-02	1.44	7
					2e-05	4.2e-02	1.37	8
					3.5e-05	7.2e-02	1.14	9
					4.8e-05	1.0e-01	1.00	10
					0.00016	3.3e-01	0.48	11
					0.00016	3.3e-01	0.48	12
					0.00019	3.9e-01	0.41	13
					0.00022	4.6e-01	0.34	14
					0.00035	7.2e-01	0.14	15

3. Motif assembly

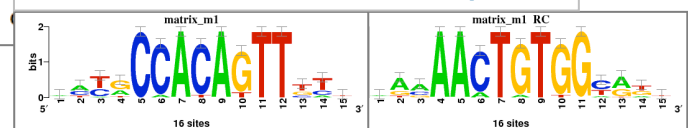
```

;assembly # 1
; align
tgccac.....
.gccaca.....
..ccacag...
...cacagt..
....acagtt.
.....cagttg
tgccacagttg

seed: ccacag 7 words
rev_cpl score
.....gtggca 0.48
....tgtggc. 3.24
....tgtgtg. 1.37
...ctgtgg.. 8.00
..actgtg... 6.45

```

4. Extract matrix from motifs in sequence



Bruno André
(ULB, Bruxelles,
Belgium)



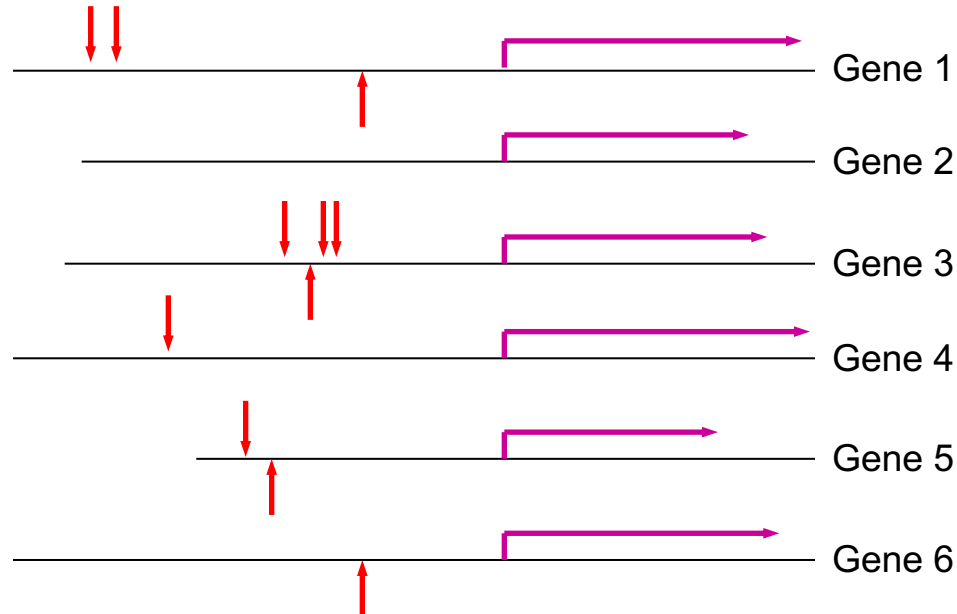
Julio Collado-Vides
(CCG, Cuernavaca –
Mexico)



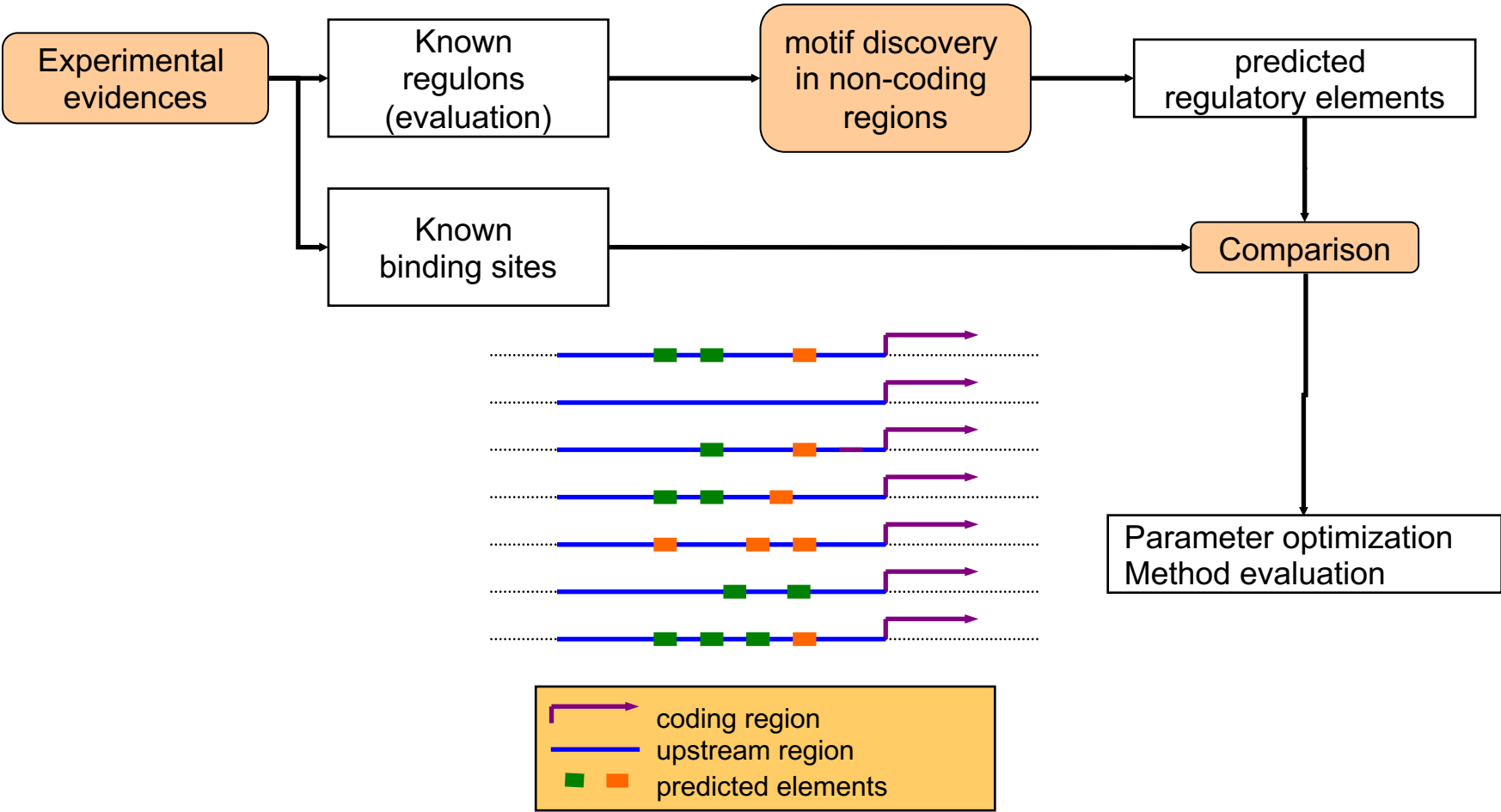
- van Helden, J., Andre, B. and Collado-Vides, J. (1998). Extracting regulatory sites from the upstream region of yeast genes by computational analysis of oligonucleotide frequencies. J Mol Biol 281, 827-42.
- van Helden, J., Andre, B. and Collado-Vides, J. (2000). A web site for the computational analysis of yeast regulatory sequences. Yeast 16, 177-87.
- van Helden, J., Rios, A. F. and Collado-Vides, J. (2000). Discovering regulatory elements in non-coding sequences by analysis of spaced dyads. Nucleic Acids Res 28, 1808-18.

Detection of over-represented motifs

- Knowing that a set of genes are co-regulated, one can expect that their upstream regions contains some regulatory signal.
- This signal is likely to be more frequent in the upstream regions of the co-regulated genes than in a random selection of genes.
- To discover signals responsible for the co-regulation of a group of genes, we can detect over-represented motifs in their upstream sequences.



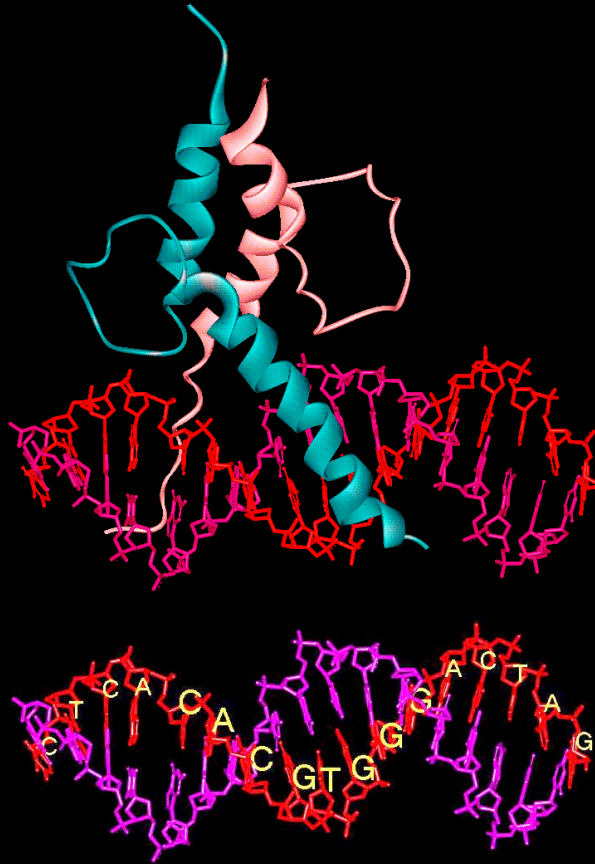
Evaluation with known regulons



Testing the performances with known regulons

- NIT
 - 7 genes expressed under low nitrogen conditions
 - DAL5, DAL80, GAP1, MEP1, MEP2, MEP3, PUT4
- MET
 - 10 genes expressed in absence of methionine
 - MET3, MET25, MET2, MET19, MET14, MET6, SAM1 SAM2, MET1, MET30, MUP3
- PHO
 - 5 genes expressed under phosphate stress
 - PHO5, PHO11, PHO8, PHO84, PHO81
- GAL
 - 6 genes expressed in presence of galactose
 - GAL1, GAL2, GAL7, GAL80, MEL1, GCY1
- ...

Interface between the yeast Pho4p protein and one of its binding sites



Background model

- To detect over-represented motifs, the observed occurrences are compared to the random expectation.
- The random expectation can be estimated according to different models
 - **Bernoulli model**, with a specific probability for each nucleotide.
 - over-simplistic to reflect biological sequence properties
 - **Markov model**, estimated from the input sequence itself
 - the order of the Markov model is restricted by the input sequence size (needs to be sufficient to obtain a reliable estimate of 3^m parameters)
 - **External background**: occurrences for the same motif in a reference data set
 - whole genome
 - Problematic :mixture of sequence types with very different properties
 - intergenic sequences
 - Include upstream and downstream sequences + “gene deserts” + heterochromatin
 - set of all upstream sequences for the organism considered

The most frequent oligonucleotides are not informative

- A (too) simple approach would consist in detecting the most frequent oligonucleotides (e.g. hexanucleotides) for each group of upstream sequences.
- This would however lead to deceiving results.
 - In all the sequence sets, the same kind of motifs are selected: AT-rich hexanucleotides.

PHO		MET		NIT		GAL	
aaaaaa tttttt	51	aaaaaa tttttt	105	aaaaaa tttttt	80	aaaaaa tttttt	47
aaaaag cttttt	15	atatat atatat	41	cttatc gataag	26	aaaaat attttt	17
aagaaa tttcct	14	gaaaaa tttttc	40	tatata tatata	22	aatata tatatt	17
gaaaaa tttttc	13	tatata tatata	40	ataaga tcctat	20	aaaatt aatttt	16
tgccaa ttggca	12	aaaaat attttt	35	aagaaa tttcct	20	aaaata tatttt	15
aaaaat attttt	12	aagaaa tttcct	29	gaaaaa tttttc	19	attttc gaaaat	13
aaatta taattt	12	agaaaa ttttct	28	atatat atatat	19	aaataa ttattt	13
agaaaa ttttct	11	aaaata tatttt	26	agataa ttatct	17	aaatat atattt	13
caagaa ttcctg	11	aaaaag cttttt	25	agaaaa ttttct	17	ataaaa ttttat	12
aaacgt acgttt	11	agaaat atttct	24	aaagaa ttcctt	16	atatta taatata	12
aaagaa ttcctt	11	aaataa ttattt	22	aaaaca tgtttt	16	atatat atatata	11
acgtgc gcacgt	10	taaaaa ttttta	21	aaaaag cttttt	15	tgaaaa ttttca	11

A more relevant criterion for over-representation

- The most frequent motifs do not reveal the motifs specifically bound by specific transcription factors.
- They merely reflect the compositional biases of upstream sequences.
- A more relevant criterion for over-representation is to detect motifs which are more frequent in the upstream sequences of the selected genes (co-regulated) than the random expectation.
- The random expectation is calculated by counting the frequency of each motif in the complete set of upstream sequences (all genes of the genome).

Estimation of word-specific expected frequencies with a Markov model

- In a Markov model, the probability to find a letter at position i depends on the residues found at the m preceding residues.
- The tables represent the transition matrices for Markov chain models of order 1 (top) and 2 (bottom).
- Expected frequencies can be estimated
 - On the basis of a set of **background sequences** (e.g. the whole set of upstream sequences of the considered organism).
 - On the basis of the **input sequence set** itself: the probability of larger words is estimated from the observed frequencies of the sub-words that compose them.

$$P(S, m) = P(S_{1,m}) \prod_{i=m+1}^L P(r_i | S_{i-m, i-1})$$

Transition matrix, order 1

	g	a	c	t
a	0.178	0.369	0.165	0.288
c	0.166	0.327	0.191	0.316
g	0.190	0.313	0.211	0.286
t	0.175	0.273	0.180	0.372

Transition matrix, order 2

	g	a	c	t
aa	0.185	0.411	0.152	0.252
ac	0.171	0.348	0.186	0.296
ag	0.193	0.337	0.201	0.269
at	0.163	0.343	0.167	0.326
ca	0.181	0.344	0.184	0.291
cc	0.168	0.313	0.198	0.321
cg	0.194	0.283	0.227	0.295
ct	0.187	0.240	0.189	0.384
ga	0.186	0.407	0.145	0.262
gc	0.180	0.331	0.194	0.295
gg	0.192	0.318	0.216	0.274
gt	0.199	0.305	0.159	0.338
ta	0.160	0.304	0.182	0.354
tc	0.151	0.313	0.192	0.344
tg	0.184	0.302	0.210	0.304
tt	0.168	0.220	0.195	0.417

Estimation of word-specific expected frequencies from a set of background sequences

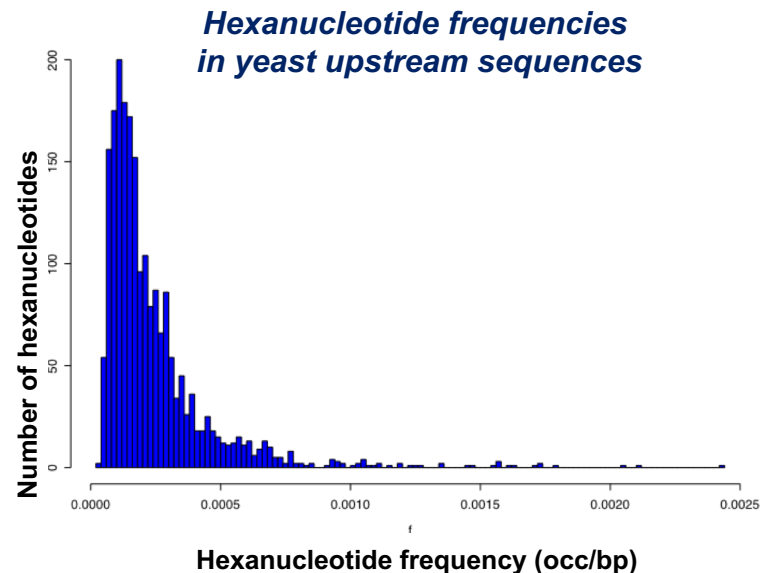
Example:

6nt frequencies in the whole set of yeast upstream sequences

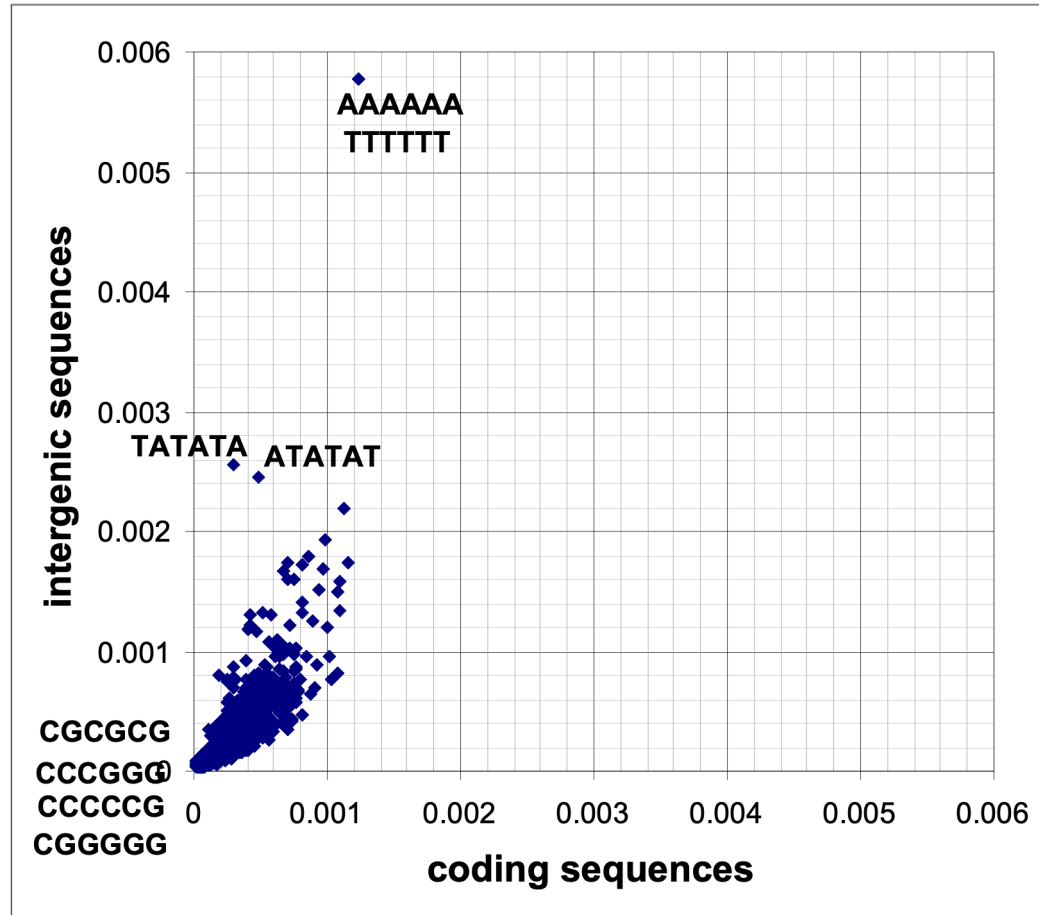
Words are grouped by pairs of reverse complements.

;seq	identifier	observed_freq	occ
aaaaaa	aaaaaa ttttt	0.00510699	14555
aaaaac	aaaaac gtttt	0.00207402	5911
aaaaag	aaaaag ctttt	0.00375191	10693
aaaaat	aaaaat atttt	0.00423577	12072
aaaaca	aaaaca tgttt	0.0019828	5651
aaaacc	aaaacc ggttt	0.00088526	2523
aaaacg	aaaacg cgttt	0.00090105	2568
aaaact	aaaact agttt	0.0014621	4167
aaaaga	aaaaga tcttt	0.00323016	9206
aaaagc	aaaagc gcttt	0.00135824	3871
aaaagg	aaaagg ccttt	0.0017849	5087
aaaagt	aaaagt acttt	0.0019035	5425
aaaata	aaaata tattt	0.00336805	9599
aaaatc	aaaatc gattt	0.00131368	3744
aaaatg	aaaatg cattt	0.00185648	5291
aaaatt	aaaatt aattt	0.00269156	7671
aaacaa	aaacaa ttggt	0.00209999	5985
aaacac	aaacac gtggt	0.00071684	2043
aaacag	aaacag ctggt	0.00096491	2750
aaacat	aaacat atggt	0.00108982	3106
aaacca	aaacca tggtt	0.00074421	2121

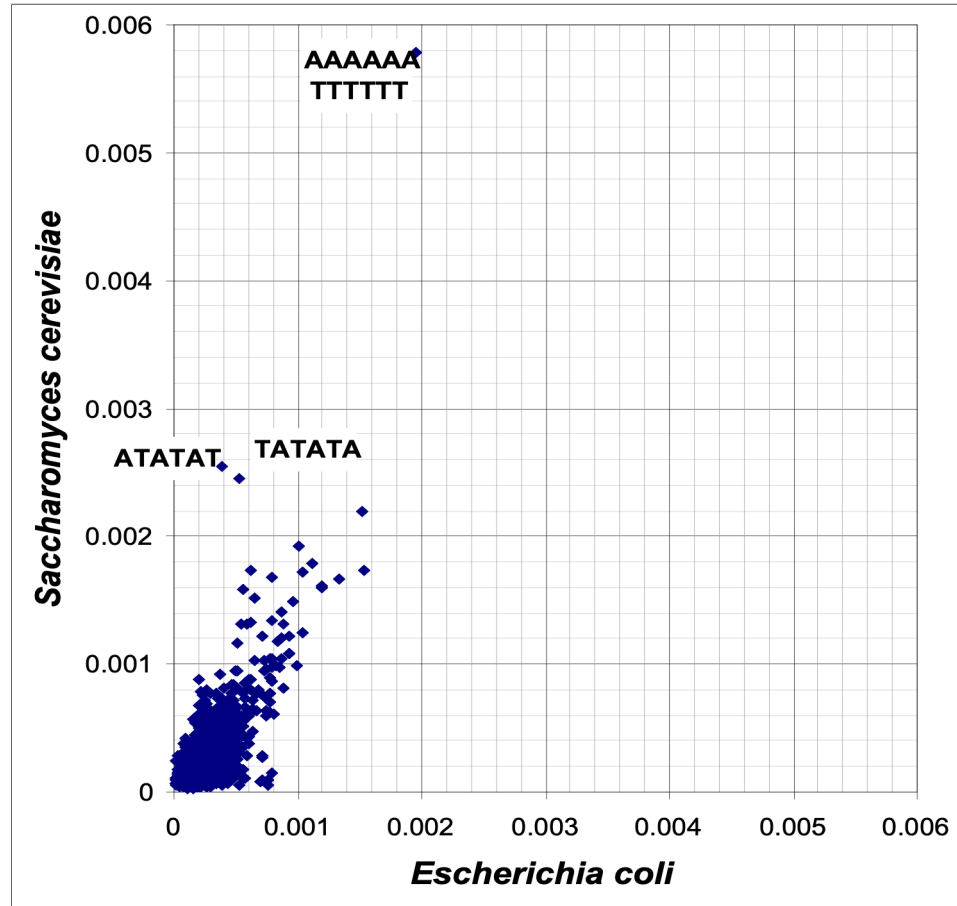
- Hexanucleotide frequencies have been measured in the whole set of 6000 yeast upstream sequences.
- Some words are very frequent, others are rare.
 - range $4.5e^{-5}$ to $1.2e^{-2}$
 - Ratio between the most frequent and less frequent hexanucleotide:
 - $\max(f)/\min(f)=268$



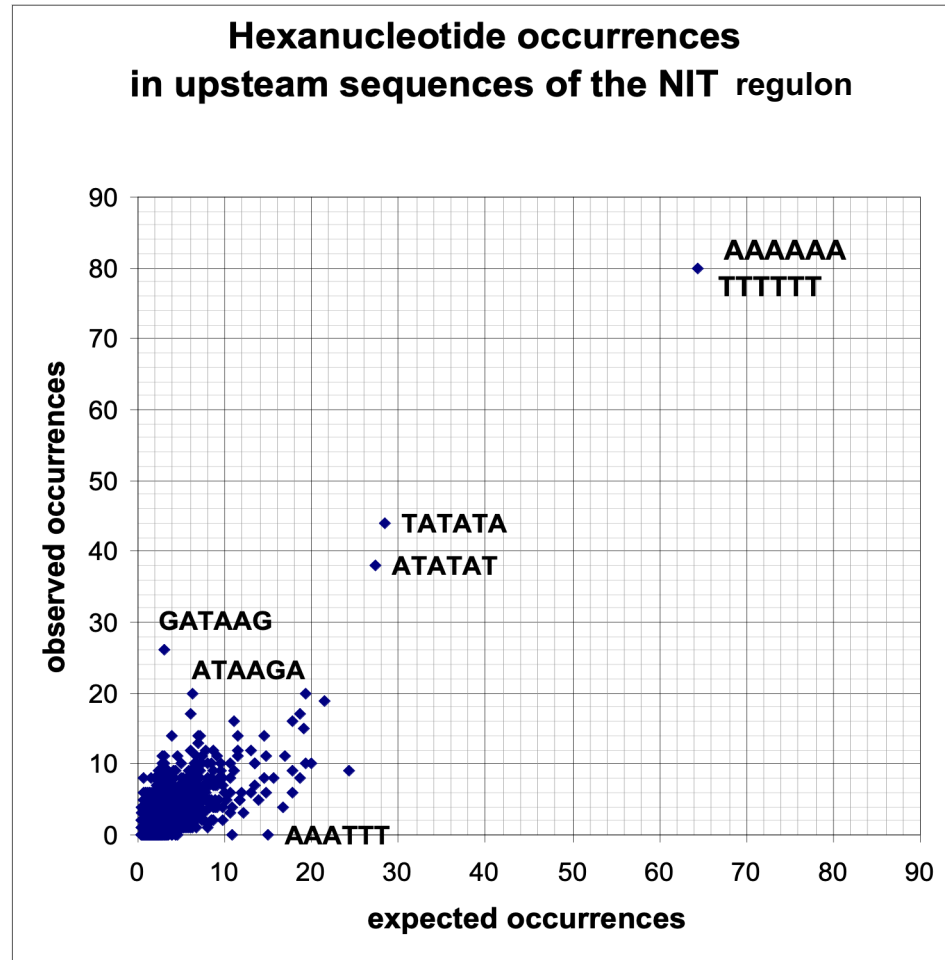
6nt frequencies differ between coding and non-coding sequences

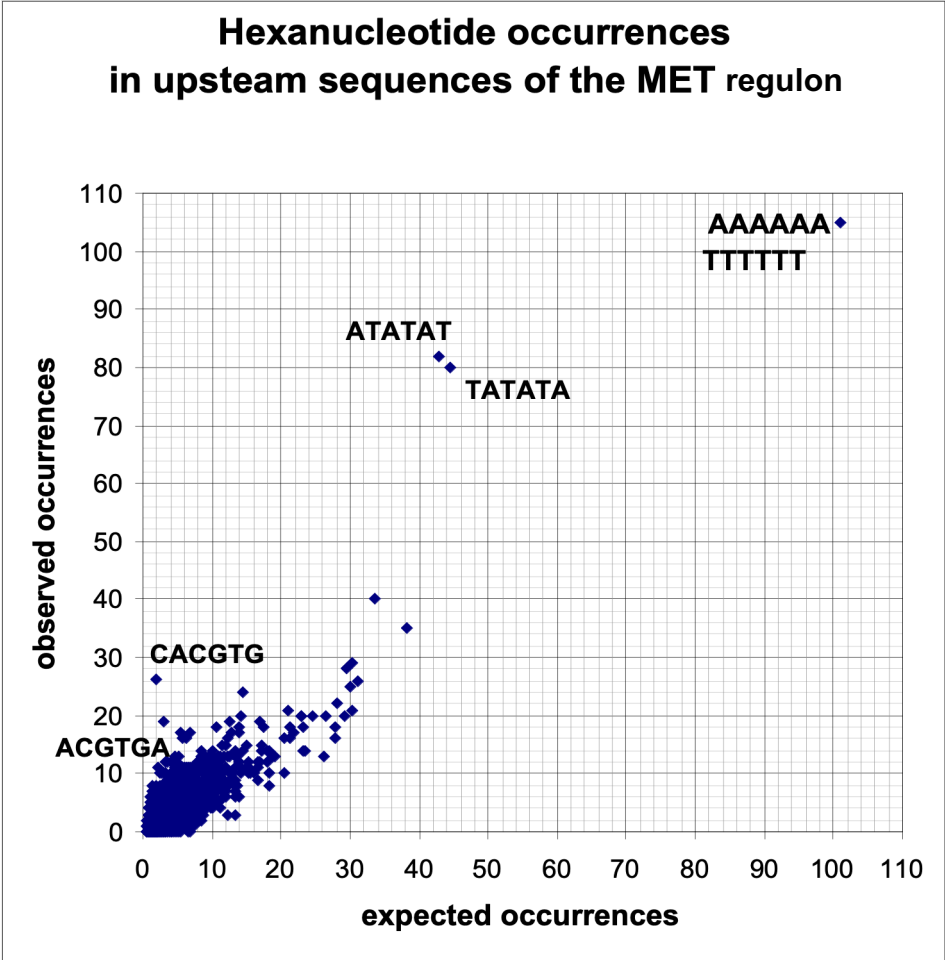


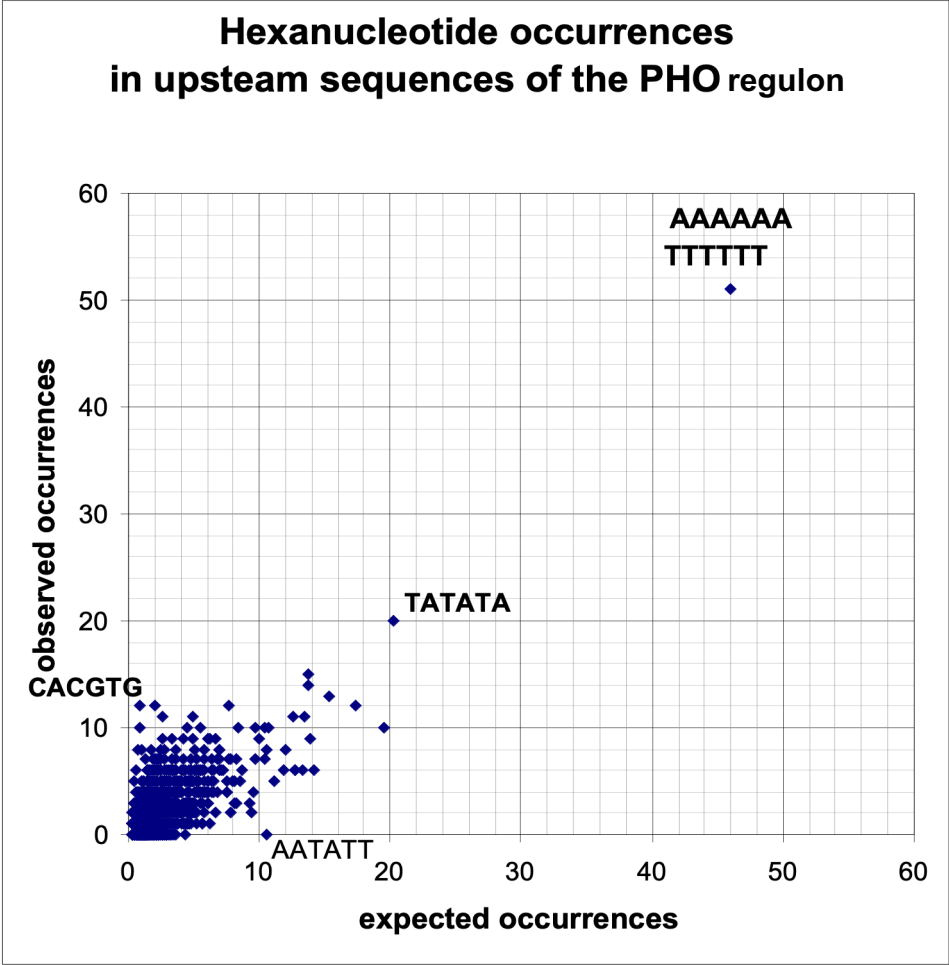
Inter-species variations in intergenic 6nt frequencies



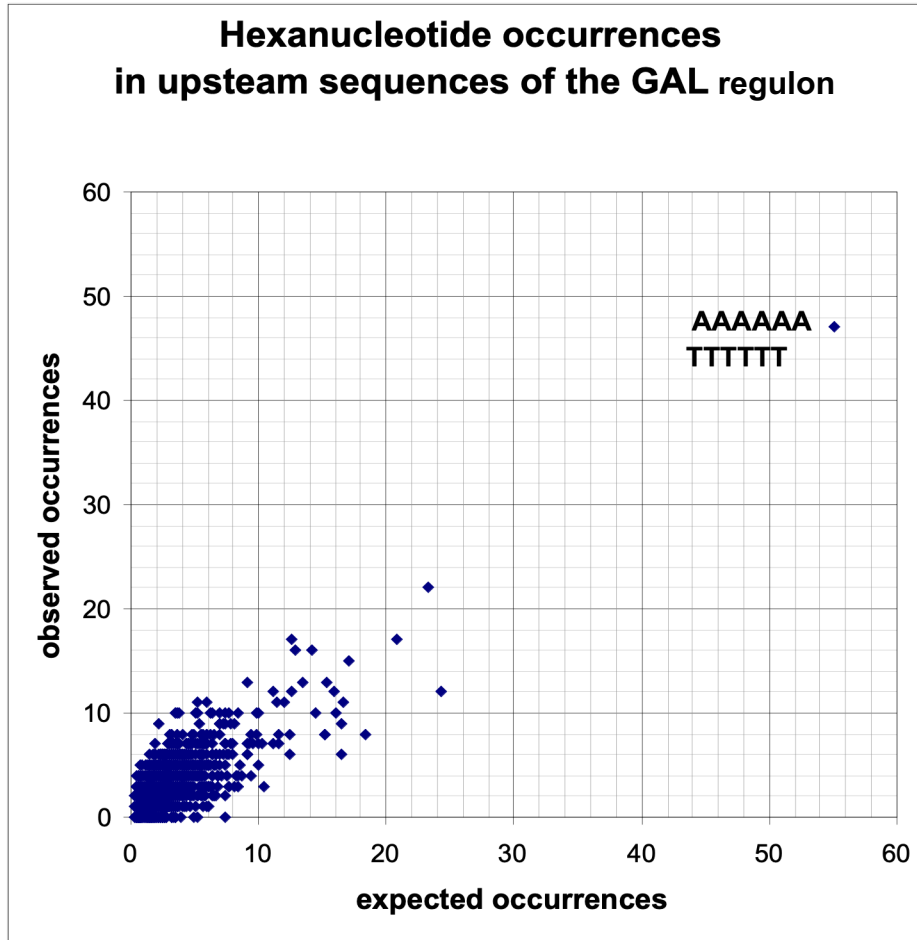
Hexanucleotide occurrences in upstream sequences of NIT genes



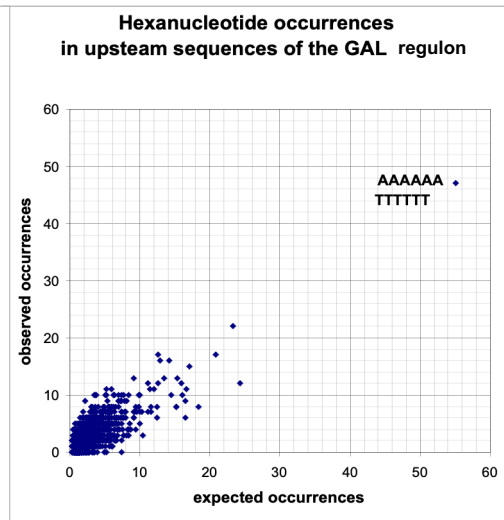
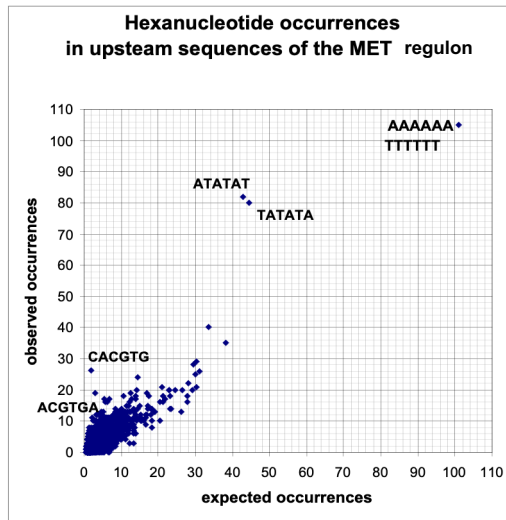
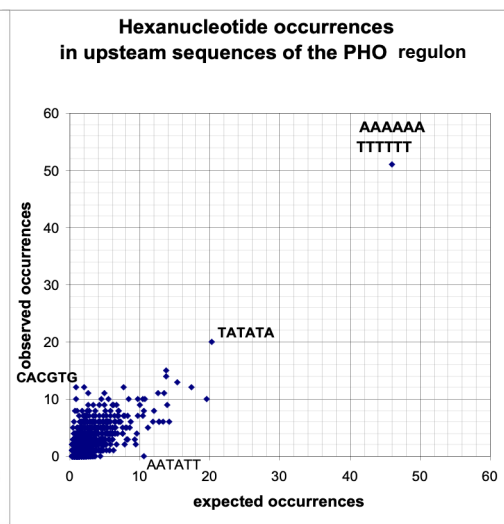
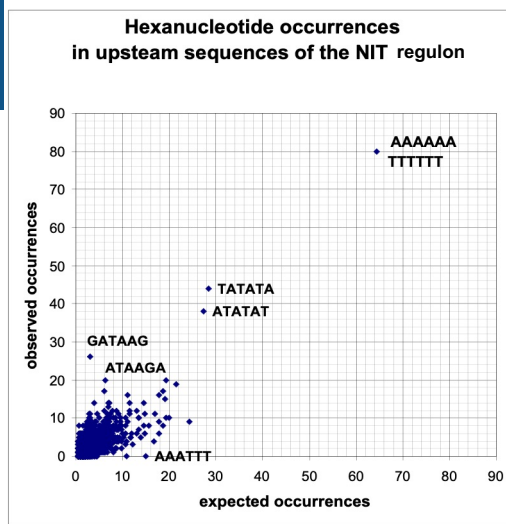




Hexanucleotide occurrences in upstream sequences of GAL genes

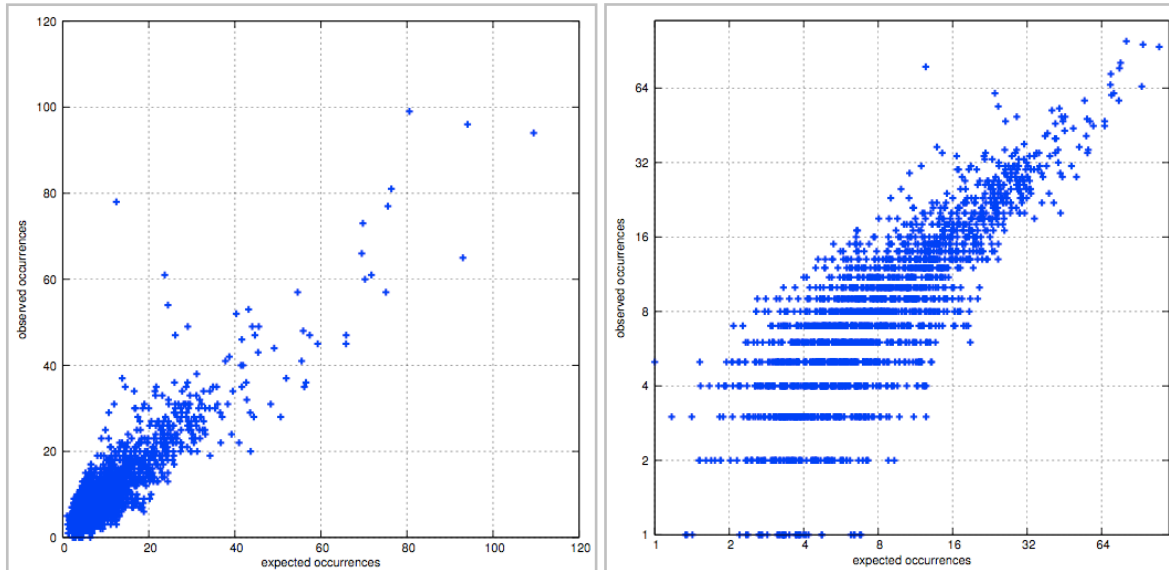


Hexanucleotide occurrences in upstream sequences of yeast regulons



Hexanucleotide occurrences in an extended NIT regulon

- We analyzed here an extended set of 41 NIT genes (taken from Godard et al., 2006).
- The number of genes affects the dispersion around the diagonal on the plot of observed versus expected occurrences.
- The signal-to-noise separation increases when more genes are analyzed.
- The logarithmic axes better emphasize the words with low expected and observed occurrences but does not allow to display words with 0 occurrences.
- Words with very low expected frequencies are sensitive to low-number fluctuations. For such cases, the observed/expected ratio is misleading (e.g. $\text{exp}=1$, $\text{obs}=4$).



Scoring statistics

- Several scoring statistics have been used to assess the statistical significance of word over-representation
- Observed/expected ratio
 - **Never use this statistics !**
 - The ratio can be misleading, because it over-emphasizes the motifs with a very low number of expected number of occurrences
 - Example:
 - $x_{obs}/x_{exp} = 10/1$ is quite significant, but $x_{obs}/x_{exp} = 1/0.1$ is not.
- Log-likelihood ratio
 - $LLR = F_{obs} * \log(F_{obs}/F_{exp})$
- Z-score (Matthieu Blanchette)
 - $Z\text{-score} = (x_{obs} - x_{exp})/s_x$
 - Only valid for very large sequences ($exp \gg 10$ for each word)
- Poisson (Andreas Wagner)
- Compound Poisson (Sophie Schbath)
- Binomial (Jacques van Helden)

Scoring statistics - Binomial

- Advantages
 - Allows to estimate a P-value.
 - Appropriate for small sequence sets, where some words have a very low expected number of occurrences (<1).
 - Allows to detect over- and under-representation.
- Weaknesses
 - Bias for self-overlapping words (but this can be circumvented by preventing the counting of overlapping occurrences).
 - Assumes that sequence length is much larger than word length

- Probability to observe exactly x occurrences

$$P(X = x) = \frac{T!}{x!(T-x)!} p^x (1-p)^{T-x}$$

- Probability to observe at least s occurrences

$$P(X \geq x) = \sum_{i=x}^T \frac{T!}{i!(T-i)!} p^i (1-p)^{T-i}$$

Where

x = observed occurrences

$T = \text{Sum}_{i=1 \rightarrow n} (L_i - k + 1)$ = number of possible positions for a word of length k in a sequence of n sequences of length L_i

p = word probability

Hexanucleotide analysis in sequences upstream of the NIT regulon

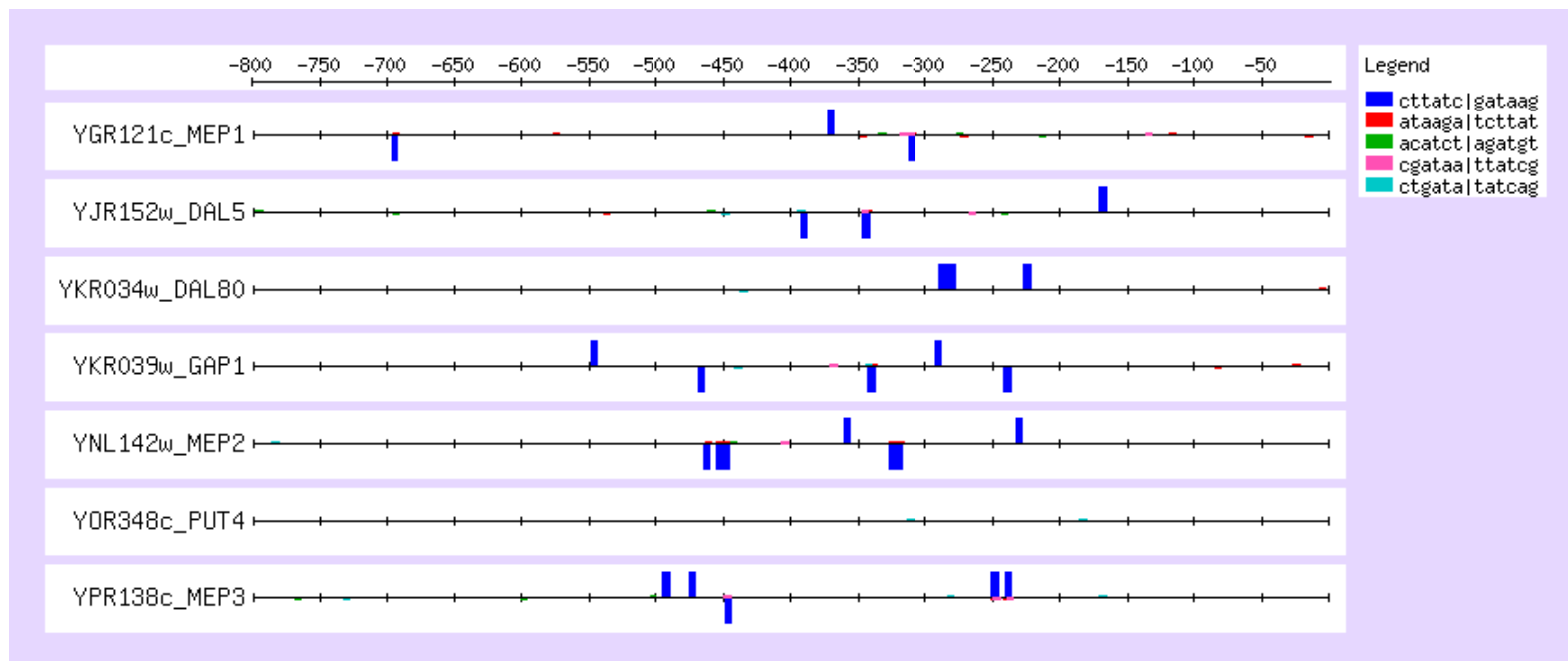
Sequence	Reverse complement	prior 6-mer probability	occ	exp occ	P-value	E-value	sig	ovl_occ	matching sequences
..TTATCG	CGATAA..	0.0004789660232	10	2.32	0.00016	3.2e-01	0.49	0	5
..TTATCT	AGATAA..	0.0009460577158	15	5.26	0.00038	7.9e-01	0.10	2	7
.CTTATC.	.GATAAG.	0.0005355636681	24	2.98	2.2e-14	4.5e-11	10.35	2	6
TCTTAT..	..ATAAGA	0.0009408463656	18	5.24	9.8e-06	2.0e-02	1.69	20	6
ACATCT	AGATGT	0.0005503959726	11	3.06	0.00035	7.2e-01	0.14	00	4
CTGATA	TATCAG	0.0005578121247	11	3.10	0.00039	8.0e-01	0.10	0	6

Genes
Known motif
Factors

DAL5, DAL80, GAP1, MEP1, MEP2, MEP3, PUT4
GATAAg
Gln3p; Nil1p; Gzf3p; Uga43p

Feature-map of discovered motifs - NIT regulon

- Typical features of yeast GATA-boxes
 - Multiple occurrences per sequences.
 - Occurrences generally appear clustered (at least two with a spacing of 0-60bp).
 - This probably stimulates synergic effects.
- Remark: PUT4 promoter does not contain a single instance of the significant hexanucleotides



Hexanucleotide analysis of the PHO regulon

Sequence	prior 6-mer probability	occ	exp occ	P-value	E-value	sig	matching sequences
.....CGTGGG	0,00013	5	0,5	0,00021	4,30E-01	0,36	3
....ACGTG <u>c</u> .	0,00021	9	0,8	2,50E-07	5,20E-04	3,29	5
....ACGTGG.	0,00018	7	0,7	9,00E-06	1,90E-02	1,73	5
...CACGTG..	0,00012	6	0,5	8,90E-06	1,90E-02	1,73	5
.cgCACG....	0,00013	6	0,5	1,40E-05	2,90E-02	1,54	5
ctgCAC...	0,00024	8	1,0	7,80E-06	1,60E-02	1,79	4
....ACGT <u>TTT</u> .	0,00061	10	2,4	0,00019	3,90E-01	0,41	5
...CACGT <u>T</u> ..	0,00030	7	1,2	0,00024	5,00E-01	0,3	5
tgccaa	0,00048	12	1,9	7,40E-07	1,50E-03	2,81	4

Genes

Known motifs

Factors

PHO5, PHO8, PHO11, PHO84, PHO81

CACGTGGG

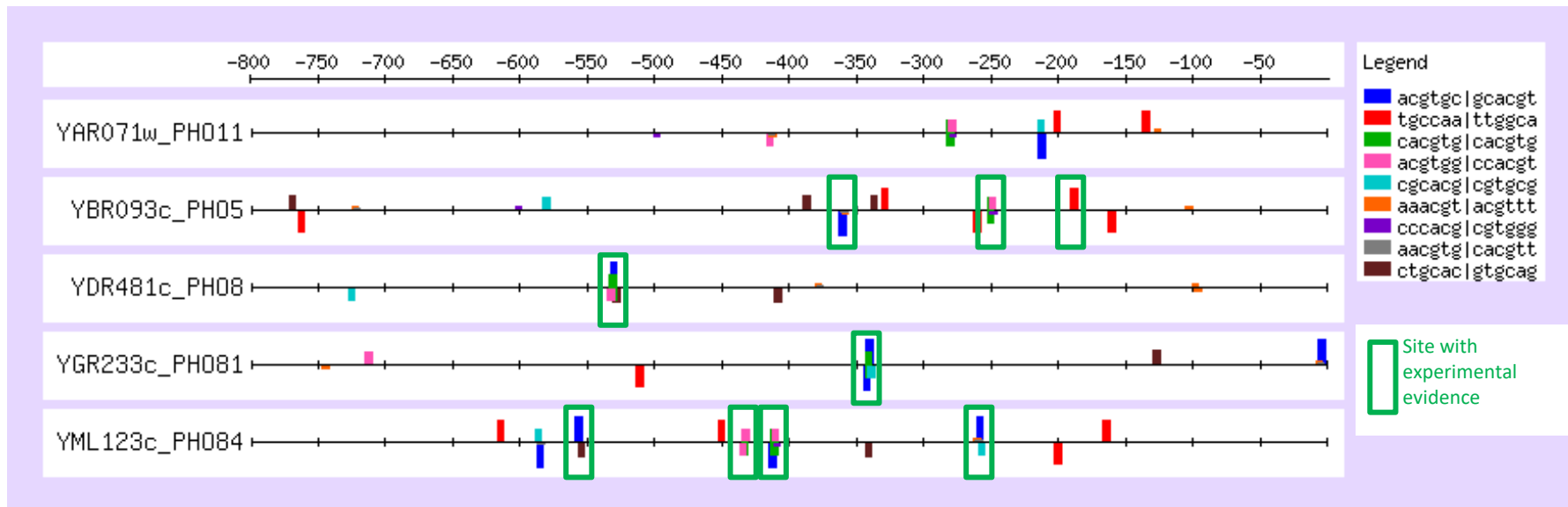
CACGTTTT

Pho4p (high affinity)

Pho4p (medium affinity)

Feature-map of over-represented k-mers – PHO regulon

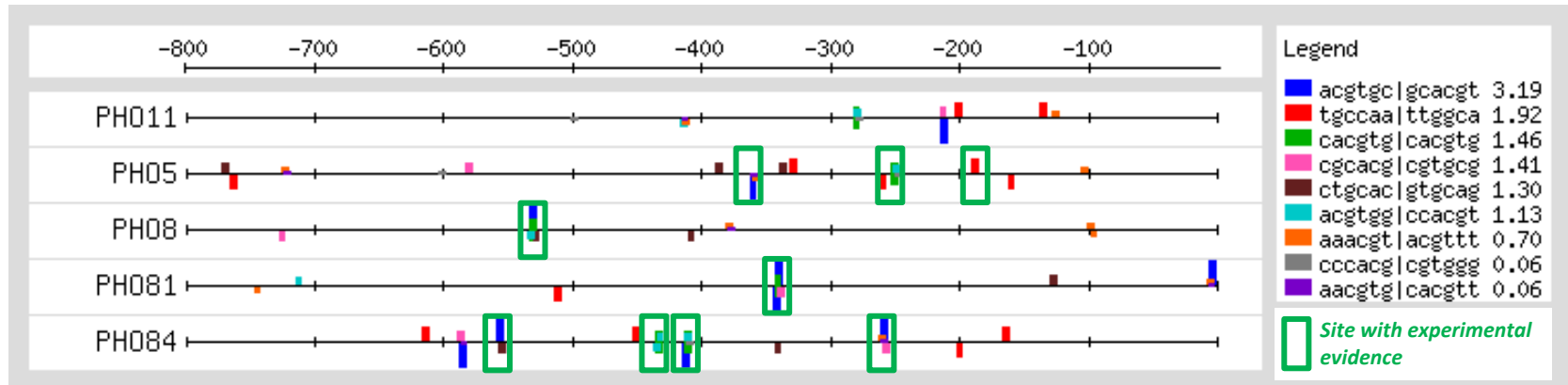
- The feature map provides a convenient representation of the location of over-represented k-mers
 - Each colour represents one over-represented k-mer
 - Box height = k-mer significance
 - Clusters of mutually overlapping words suggest the presence of TFBS wider than 6 bp.
- Green rectangles indicate the positions of experimentally proven sites
 - For PHO11, no site is documented, we can thus not check the predictions.
 - For the other genes, the proven sites are detected as clusters of overlapping words



Feature-map of over-represented k-mers - PHO regulon

- The feature map provides a convenient representation of the over-represented k-mers
 - Each colour represents a different k-mer
 - Green boxes highlight sites with experimental evidence
- Green boxes highlight sites with experimental evidence
 - PHO11: sites at approximately -350, -250, and -150
 - PHO5: sites at approximately -450, -350, -250, and -150
 - PHO8: site at approximately -550
 - PHO81: sites at approximately -450 and -350
 - PHO84: sites at approximately -550, -450, -350, and -250

Reproducibility
 Same analysis as in 1998 :
 retrieve-seq → oligo-analysis → dna-motif → feature-map
 ran on RSAT server (<http://rsat.eu>) on Jan 12, 2025
 Main change: background color



Hexanucleotide analysis of the MET regulon

Sequence	exp freq	occ	exp occ	P-value	E-value	sig	matching sequences
. . ACGTGa	0.00033	13	2.9	1.00E-05	2.20E-02	1.67	9
. CACGTG .	0.00012	13	1.0	6.90E-11	1.40E-07	6.84	9
tCACGTG .	0.00033	13	2.9	1.00E-05	2.20E-02	1.67	9
tCACGTGa	consensus						
. . . . TGTGGc	0.00027	10	2.3	1.50E-04	3.20E-01	0.49	7
. . . CTGTGG .	0.00022	11	1.9	4.30E-06	8.90E-03	2.05	8
. . aCTGTG . .	0.00036	12	3.1	9.90E-05	2.10E-01	0.69	9
. aaCTGT . . .	0.00063	17	5.4	4.90E-05	1.00E-01	0.99	11
aaaCTG	0.00074	17	6.4	0.00037	7.60E-01	0.12	11
aaaCTGTGGc	consensus						
gcttcc	0.00039	12	3.4	0.00021	4.50E-01	0.35	7

Genes

Known motifs

Factors

SAM2, MET6, MUP3, MET30, MET3, MET14, MET1, SAM1, MET17, ZWF1, MET2

TCACGTG

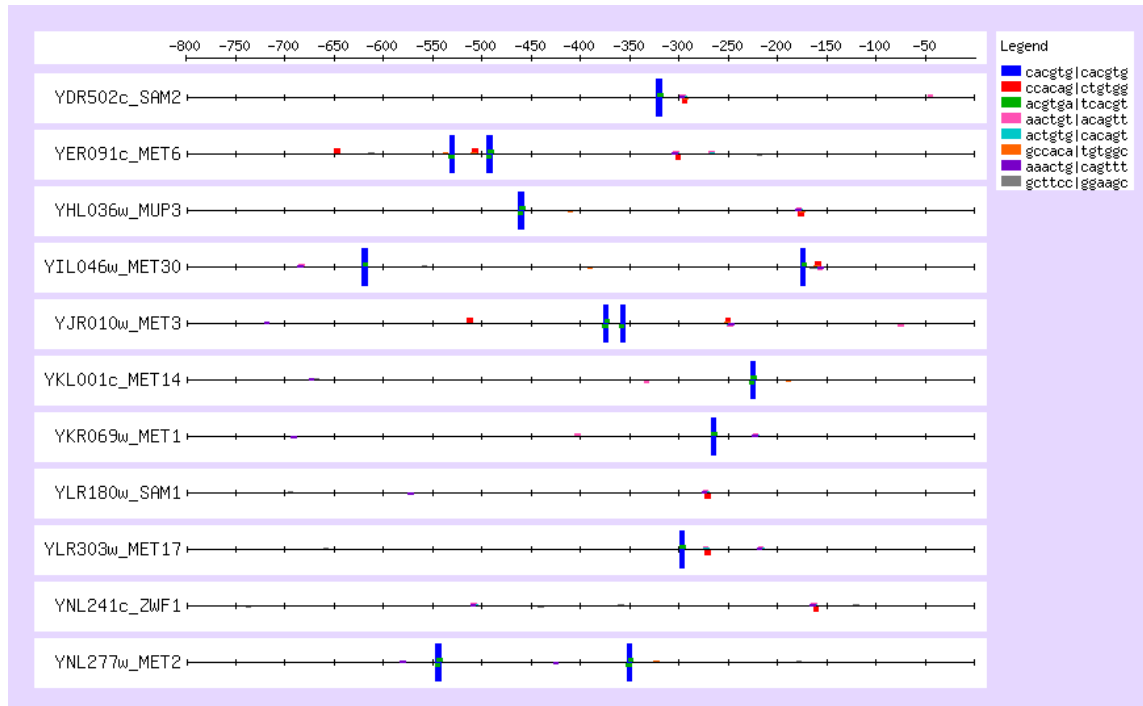
Cbf1p/Met4p/Met28p

AAACTGTGG

Met31p; Met32p

Feature-map of discovered motifs - MET regulon

- Two distinct motifs (combinations of words) are apparent.
 - blue-green TCACGTGA Met4p/Met28p/Cbf1p
 - red-violet AAAGTGTG Met31p; Met32p
- Multiple clustered motifs are sometimes found, but not always.



Expected frequency calibration

- The results of string-based motif discovery depend drastically on the choice of a background model.
- Taking the MET regulon as example
 - With 6nt calibration in intergenic sequences, the Met4p binding site appears at rank 1, and Met31p at rank 3
 - With equiprobable nucleotides, Met4p only appears are rank 20, and Met31p at rank 32. In other terms, they will never be considered as the most interesting motifs
 - With a single-nucleotide calibration, the Met4p appears at rank 4 and Met31p at rank 13. The first motif would thus have been easily detected, but not the second one.

pattern	rev compl	Background model		
		intergenic	Bernoulli	equiprobable
atcacg....	cgtgat	9	44	139
gtcacg....	cgtgac	5	34	266
.tcacgt...	...acgtga.	2	4	20
..cacgtg..	..cacgtg..	1	3	23
...acgtga.	.tcacgt...	2	4	20
....cgtgac	gtcacg....	5	34	266
....cgtgat	atcacg....	9	44	139
gccaca....	tgtggc	7	17	164
.ccacag...	...ctgtgg.	3	13	99
..cacagt..	..actgtg..	6	21	75
...acagtt.	.aactgt...	4	19	32
....cagttt	aaactg....	10	18	33
gcttcc	ggaagc	8	10	77

Effect of oligonucleotide size on the significance

Family	Pattern	oligonucleotide length					
		4	5	6	7	8	9
NIT	aGATAAGa	1.8	4.1	9.1	4.6	0.9	-
MET	gTCACGTG	4.4	4.1	7	8.2	3.2	-
	AAACTGTGg	1.5	2.3	1.6	4.8	5.2	4.9
PHO	CACGTggg	4.7	8.4	4.4	4.3	4.3	-
	aTGCCAA	2.6	1.5	2.6	0.6	-	-
	CTGCAC	-	-	1.7	-	-	-
INO	CAACAAG	2.9	2.1	3.7	1.3	-	-
	cCATGTGAA	-	-	2.7	3.2	6.4	0.4
PDR	tCCGTGGa	1.5	3.3	7.4	6.9	4.2	1.4
	tCCGCGga	6.9	7.1	4.5	5.6	1.8	1
GCN4	GCNgtGACTCa	5.4	8.8	8.2	7.7	4.7	-
	CAGCGGa	3.3	3.5	4	0.6	-	-
YAP	CATTACTAA	-	-	1	2.3	2.1	3.2
	cCGTTCC	0.1	0.5	3.3	0.3	-	-
YAP (400bp)	aTTACTAA	-	-	0.7	4.5	2.5	3.5
	cCGTTCC	0.8	0.5	2.4	0.7	0.2	-
TUP	gtGGGGta	10.1	9	8.6	5.6	3	-
	catAGGCAC	3.3	3.3	4.3	2.6	3.3	1.7

oligo-analysis results with known regulons (sig > 1)

Family	Factor	DNA-binding Domain	Known motifs	oligont	reverse oligont	score
NIT	GATA factors	Zn finger	GATAAG	TCTTATCT	AGATAAGA	20.0
MET	Cbf1p/Met4p/Met28p Met31p, Met32p	bHLH/bLZ/bLZ Zn finger	TCACGTG	CACGTGAT	ATCACGTG	9.0
			AAAACTGTGG	CACGTGAC	GTCACGTG	9.0
				AACTGTGGCG	CGCCACAGTT	3.6
PHO	Pho4p (high affinity) Pho4p (medium affin.)	bHLH bHLH	GCACGTGGG	CCCACGTGCG	CGCACGTGGG	4.4
			GCACGTTTT	AAACGTGCG	CGCACGTTT	4.4
				TGCCAA	TTGGCA	2.6
				CTGCAC	GTGCAG	1.8
PDR	Pdr1p, Pdr3p	Zn ₂ Cys ₆ binuclear cluster	tytCCGYGGary	TCCGTGGAA	TTCCACGGA	7.4
				TCCGCGG	CCGCGGA	4.5
GCN4	Gcn4p	bZip	RRTGACTCTTT	ATGACTCA	TGAGTCAT	8.5
				AGTGACTCA	TGAGTCACT	8.5
				ATGACTCT	AGAGTCAT	8.5
				ATGACTCC	GGAGTCAT	8.5
				ATGACTA	TAGTCAT	3.8
				CCGCTG	CAGCGG	3.7
				GCCGGT	ACCGGC	1.3
INO	Ino2p/Opi1p	bHLH/leucine zipper	CATGTGAAT	CAACAACG	CGTTGTTG	3.8
				CAACAAG	CTTGTTG	3.8
				TTCACATG	CATGTGAA	2.8
HAP 2/3/4	Hap2/3/4/5p		CCAAY	AGAGAGA	TCTCTCT	2.8
GAL4	Gal4p	Zn ₂ Cys ₆ binucl. cluster	CGGn ₁₁ CCG	no significant pattern		

Hexanucleotide analysis of the GAL regulon

- With the GAL regulon, the program returns a single motif.
 - The significance of this motif is very low.
 - This level of significance is expected at random ~ once per sequence set.
 - This can be considered as a negative result: the program did not detect any really significant motif.
- Why did the program fail to discover the GAL4 motif ?

Sequence	exp freq	occ	exp occ	P-value	E-value	sig	matching sequences
agacat	0.00044	9	2.1	0.00033	0.69	0.16	4

Genes

Known motifs

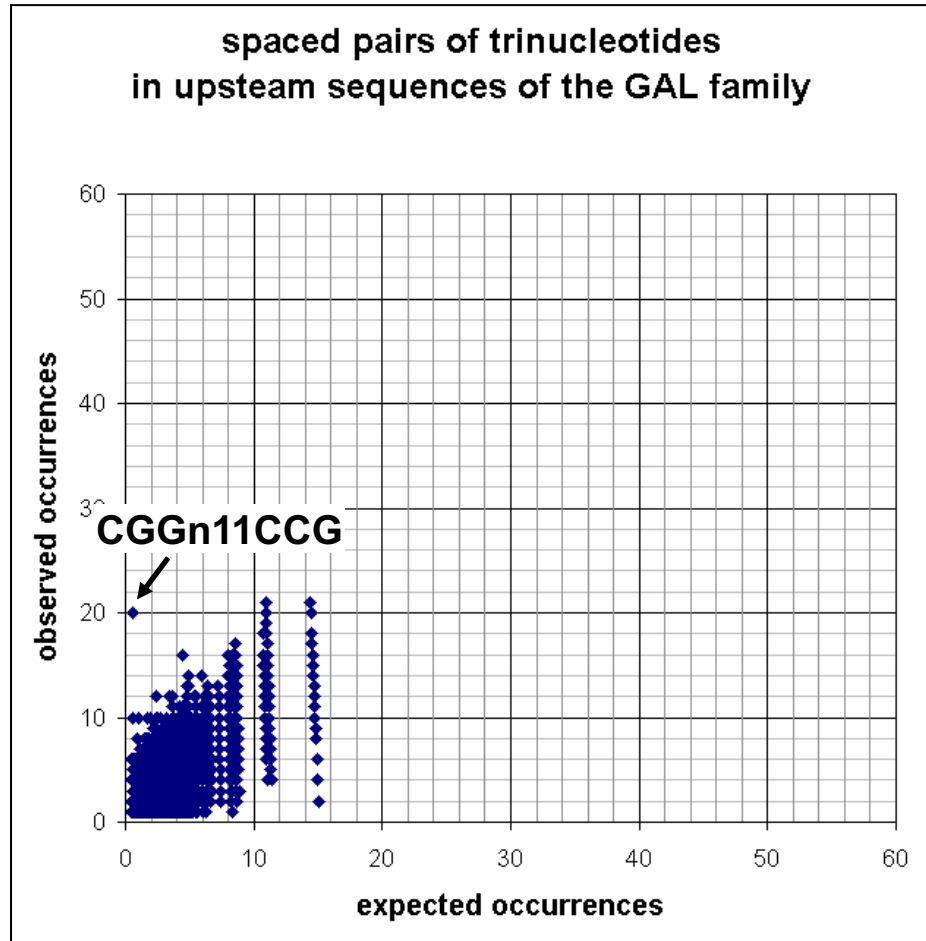
CGGn₅wn₅CCG

GAL1, GAL2, GAL7, GAL80, MEL1, GCY1

Factors

Gal4p

Occurrences of 3nt dyads in the GAL regulon



Dyad analysis of the GAL regulon

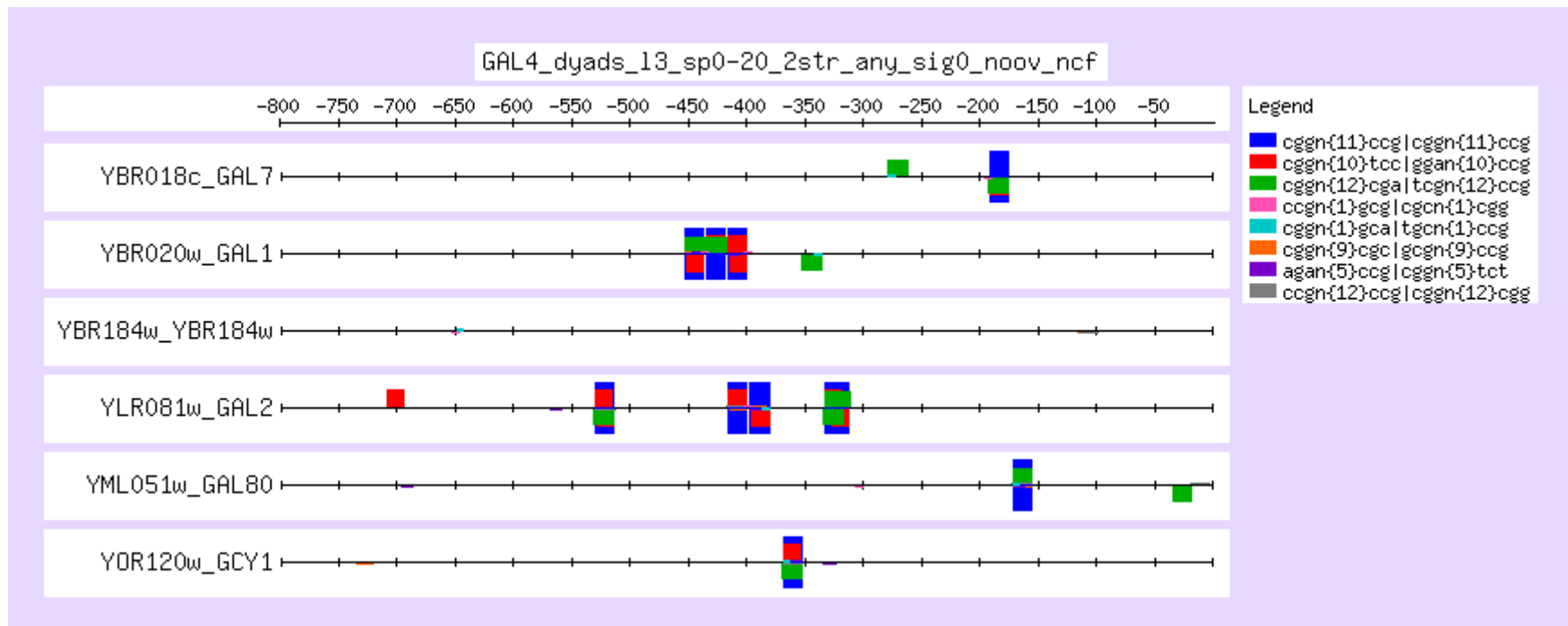
Sequence	exp freq	obs occ	exp occ	P-value	E-value	sig
..GGa.....CCG.	0.00006	10	0.5	2.70E-10	1.20E-05	4.92
.CGG.....Cga	0.00006	10	0.5	4.80E-10	2.10E-05	4.68
.CGG.....CCG.	0.00007	20	0.6	2.10E-12	9.20E-08	7.03
.CGG.....tCC..	0.00006	10	0.5	2.70E-10	1.20E-05	4.92
.CGG.....cgC...	0.00004	6	0.4	5.30E-06	2.30E-01	0.64
tCG.....CCG.	0.00006	10	0.5	4.80E-10	2.10E-05	4.68
cCG.....CCG.	0.00005	6	0.4	6.40E-06	2.80E-01	0.55
yCGGa.....ckCCGa						
AGA.....CCG	0.00010	8	0.9	7.00E-06	3.10E-01	0.51
CCG.GCG	0.00005	6	0.5	9.30E-06	4.00E-01	0.39

Genes
Known motif
Factor

GAL1, GAL2, GAL7, GAL80, MEL1, GCY1
CGGn₅wn₅CCG
Gal4p

Feature-map of discovered motifs - GAL regulon

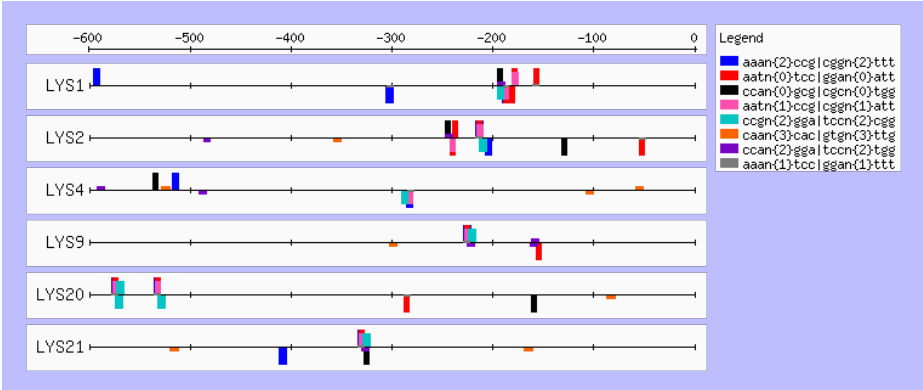
- Clusters of overlapping dyads indicates that conservation extends over 3 bp on each side of the dyad.
- Some genes, but not all, contain multiple motifs (synergic effect).



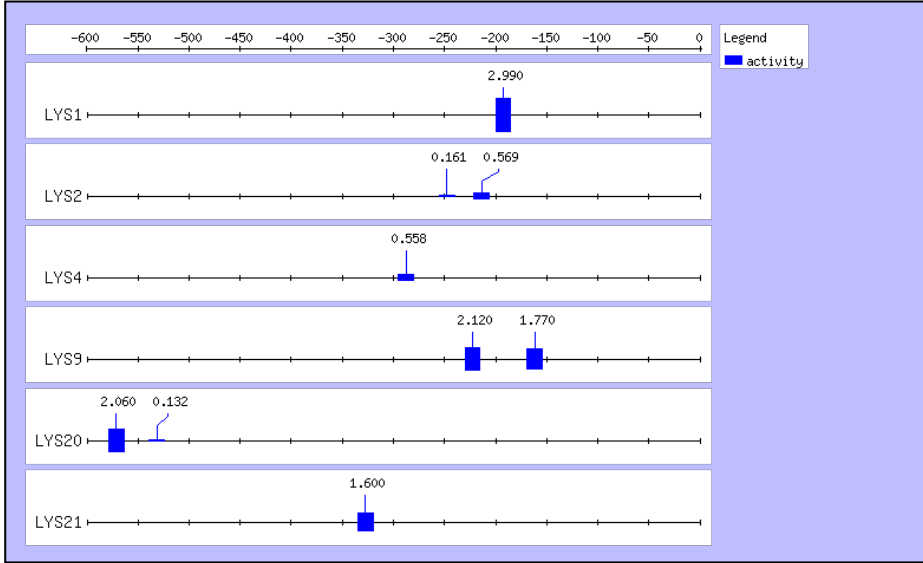
Dyad analysis: regulons of Zn cluster proteins

FACTOR	# genes	KNOWN MOTIFS	DYADS	REVERSE DYADS	SCORE
GAL4	6	CGGn ₁₁ CCG	TCGGAn ₉ TCCGG	CCGGAn ₉ TCCGA	7.8
			TCGGCGCAGAn ₄ TCCGG	CCGGAn ₄ TCTGCGCCGA	7.8
HAP1	9	CGGnnntanCGG	GGAn ₅ CGGC	GCCGn ₅ TCC	1.8
			GGGGGn ₁₂ GGC	GCCn ₁₂ CCCCC	1.4
			CCTn ₁₀ GGC	GCCn ₁₀ AGG	1.1
LEU3	5	RCCggnnccGGY	CCGn ₃ CCG	CGGn ₃ CGG	1.0
LYS	6	wwwTCCrnyGGAwww	AAATTCCG	CGGAATTT	1.9
			TCCGCTGGA	TCCAGCGGA	1.0
PDR	6	tytCCGYGGary	CTCCGTGGAA	TTCCACGGAG	6.7
			CTCCGCGGAA	TTCCGCGGAG	6.7
PPR1	3	wyCGGnnwwykCCGaw		CGGn ₆ CCG	0.5
PUT3	2	yCGGnangcgnannnCCGa	CGGn ₁₀ CCG	CGGn ₁₀ CCG	1.2
UGA3	3	aaarccgcsggcggsawt	CGGn ₁₄ AGG	CCTn ₁₄ CCG	1.7
			GCCn ₁₁ TCC	GGAn ₁₁ GGC	1.0
UME6	25	tagccgccga	TCGGCGGCTA	TAGCCGCCGA	4.9
CAT8	5	CGGnnnnnnnGGA	CGGn ₄ ATGGAA	TTCCATn ₄ CCG	6.0

Motifs discovered by dyad analysis

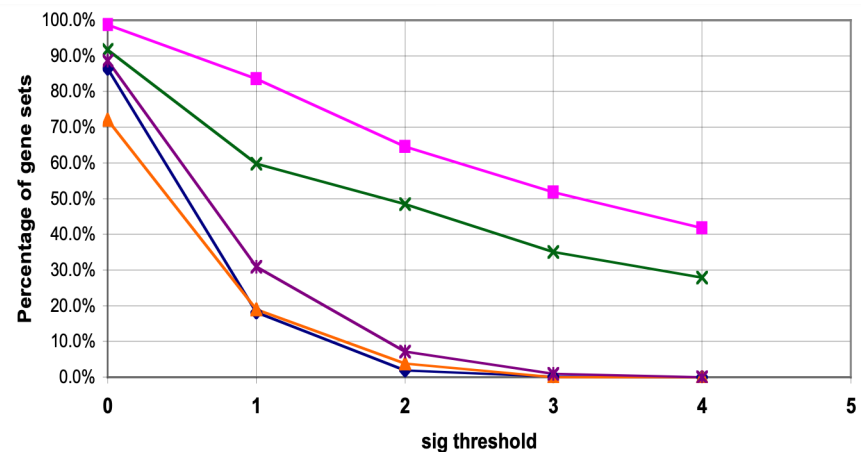
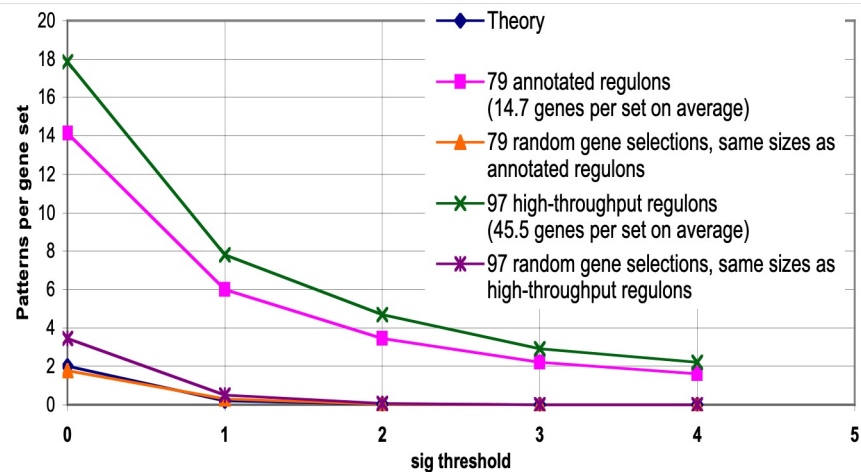


Experimental measurement of activity



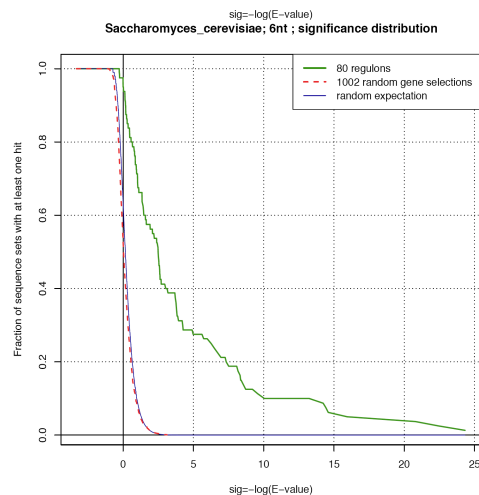
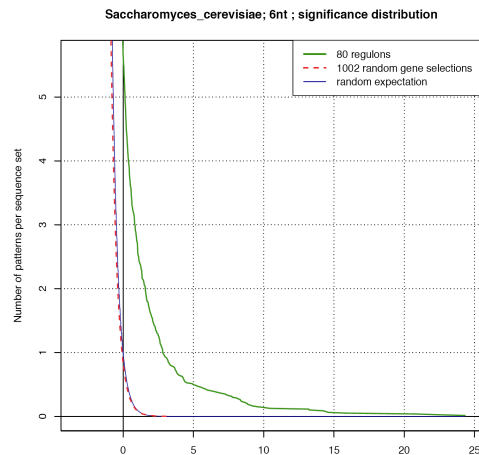
Quantitative evaluation of motif discovery results

Validation of motif discovery with yeast regulons

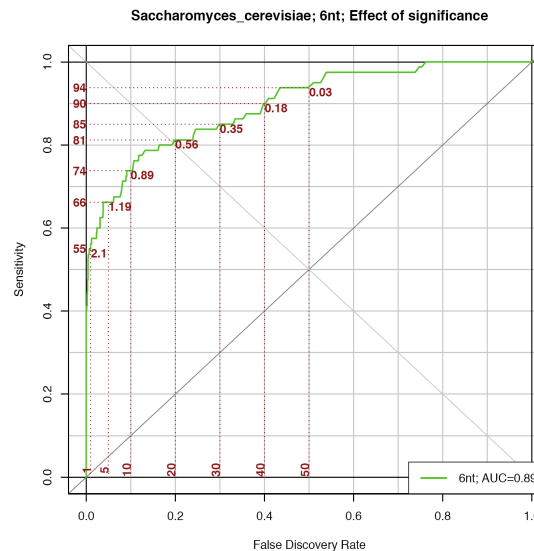


- These figures regroup over-represented motifs detected with
 - *oligo-analysis*
 - *dyad-analysis*
- Regulons were collected from TRANSFAC and aMAZE.
- All the regulons with ≥ 5 genes were analysed.
 - Significant motifs ($\text{sig} \geq 2$) are detected in 65% of the regulons.
- As a negative control, sets of random genes were analysed.
 - The rate of false positive follows pretty well the statistical expectation.

Assessment of motif discovery in yeast regulons - *Saccharomyces cerevisiae*

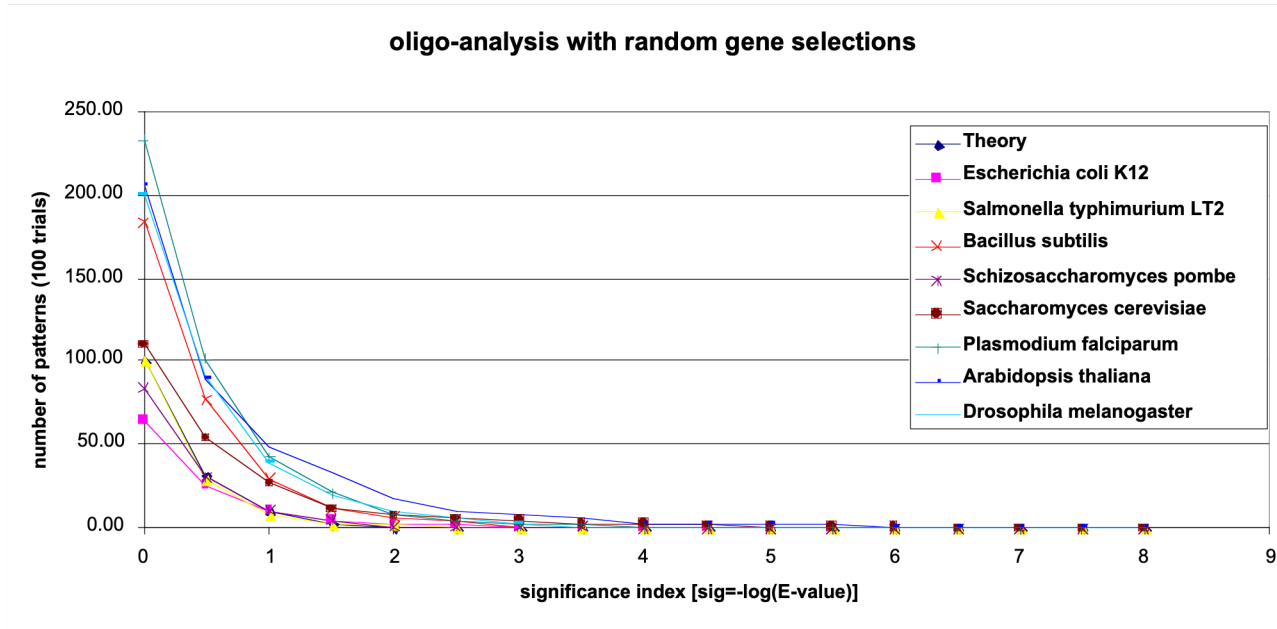


- As a control, we compare the significance of motifs discovered
 - **in regulons** (positive control)
 - **in random gene selections** (negative control)
- In the yeast *Saccharomyces cerevisiae*
 - FPR fits remarkably well the binomial P-value.
 - When the significance threshold increases,
 - sensitivity decreases (less motifs found in regulons)
 - specificity increases (less motifs in random selections)



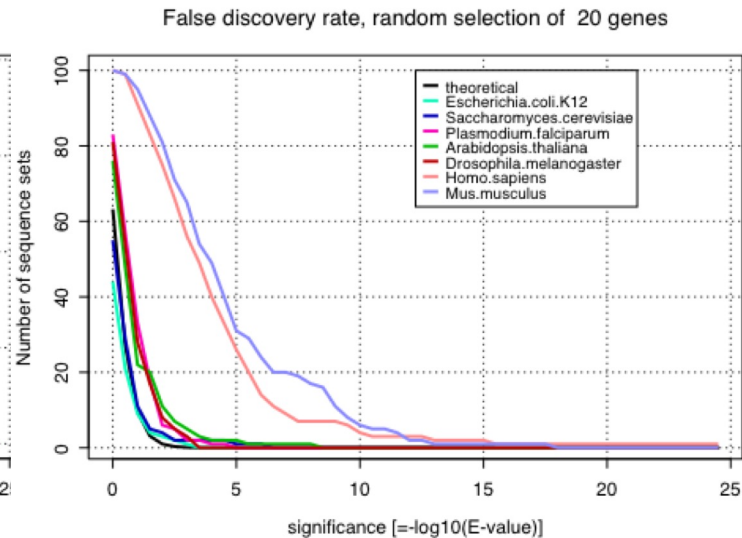
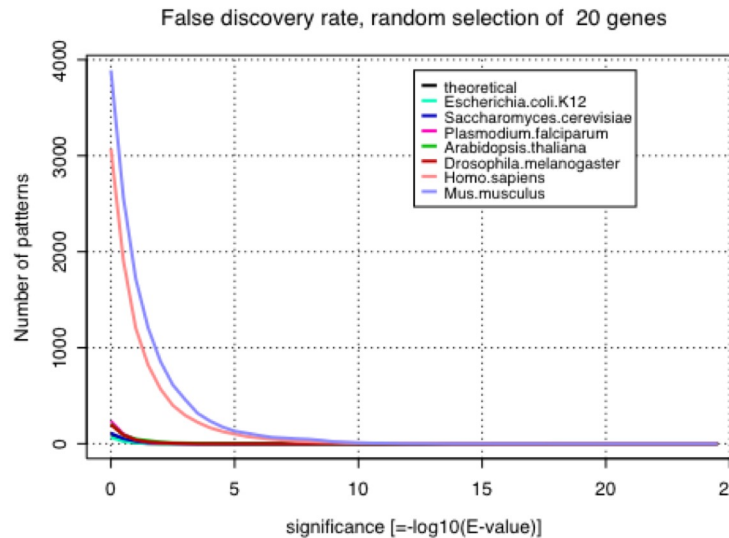
Rate of false positive in different organisms

- The analysis of random gene selections allows to evaluate the rate of false positive returned by a motif discovery program.
- The rate of false positive is good for microbes (bacteria, yeasts, ...), but increases for multicellular organisms (e.g. the fly *Drosophila*, the plant *Arabidopsis thaliana*, ...).
- The rate of false positive is also higher in the protozoan *Plasmodium falciparum* (the agent of the malaria) than in bacteria and yeast.

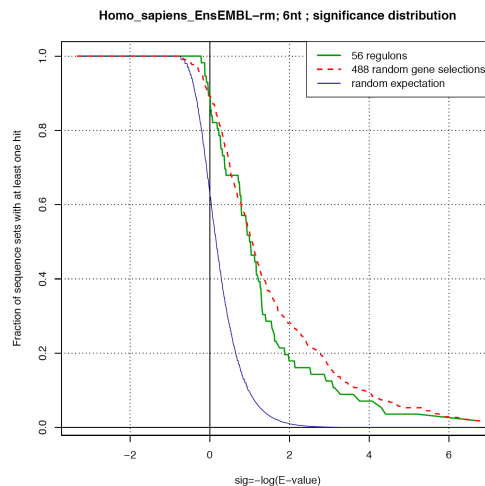
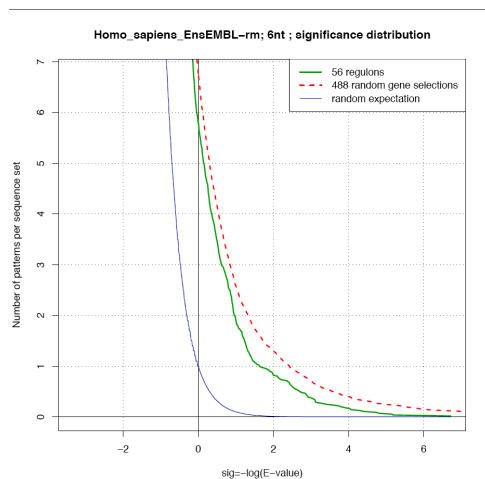


Rate of false positive in higher organisms

- The rate of false positive increases dramatically with higher organisms.
- This is likely to come from
 - a bad treatment of repetitive elements : genome-scale calibration does not account for local frequencies
 - positional heterogeneities : oligonucleotide frequencies depend on the distance from the gene
 - the higher heterogeneity of genomic sequences in these organisms (GC-rich vs AT-rich promoters)
- We are currently developing more elaborate background models to treat this problem.

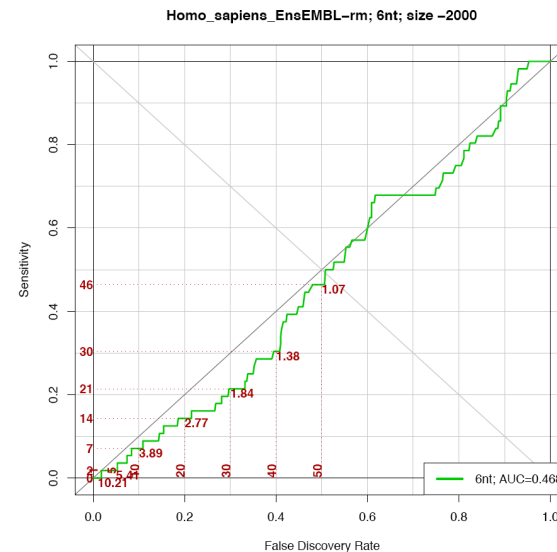


Motif significance in regulons - *Homo sapiens*



■ In *Homo sapiens*

- ❑ False positive rate (FPR) much higher than theoretical expectation
- ❑ Significance score is quite inefficient to distinguish between reliable motifs and false positives.
- ❑ Reasons:
 - Inadequacy of background models.
 - Actual TFBS are not restricted to proximal promoters.

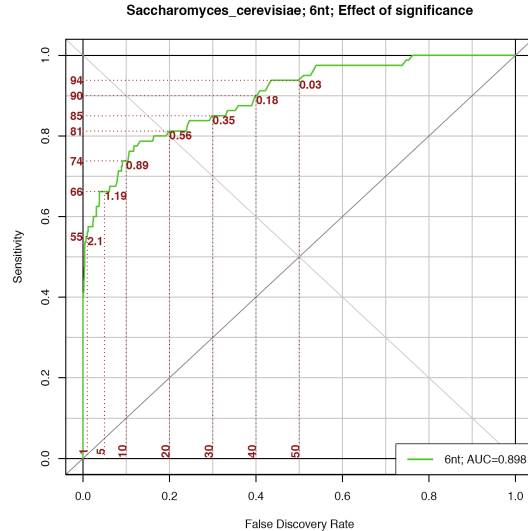




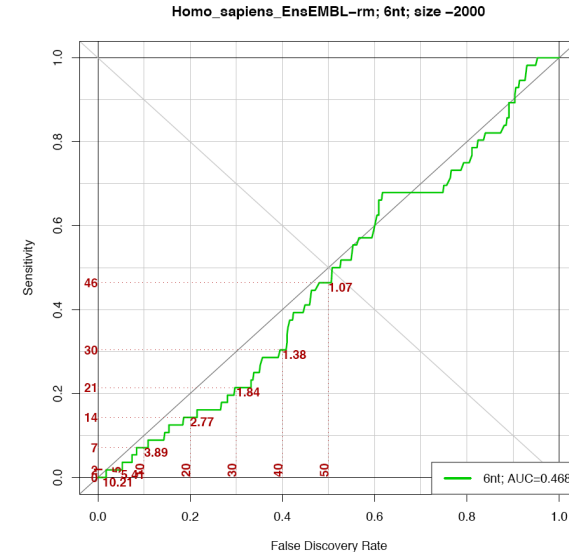
Jean Valéry
Turatsinze

Assessment of motif discovery in regulons – Yeast versus Human

- As a control, we compare the significance of motifs discovered
 - **regulons** (positive control)
 - **random gene selections** (negative control)
- In the yeast *Saccharomyces cerevisiae*
 - FPR fits remarkably well the binomial P-value.
 - When the significance threshold increases,
 - sensitivity decreases (less motifs found in regulons)
 - specificity increases (less motifs in random gene selections)



- In *Homo sapiens*
 - False positive rate (FPR) much higher than theoretical expectation.
 - Significance score cannot distinguish between reliable motifs and false positives.
 - Reasons:
 - Inadequacy of background models.
 - Actual TFBS are not restricted to proximal promoters.



String-based motif discovery: strengths

- Deterministic (not heuristic) and exhaustive
 - all possible words/dyads are tested
 - ability to return several motifs in a single run
- Speed
 - co-expression clusters are treated within seconds
- Time increases linearly with sequence set
 - Can be applied to very large sequence sets (full genomes)
 - Realistic application: ChIP-seq peaks generally cover several Mb or even tens of Mb. Such files are treated in a few minutes on a personal laptop.
- Ability to return a negative answer
 - "not a single over-represented motif in this sequence set"
 - Corollary: very low false positive rate
- Ability to detect over-represented, but also under-represented motifs
 - (e.g. restriction sites in bacterial genomes)
- Motif assembly refines the result
 - ability to detect some level of degeneracy (result contains words differing by single substitutions)
 - ability to detect motifs larger than the oligonucleotide size (result contains strongly overlapping words)

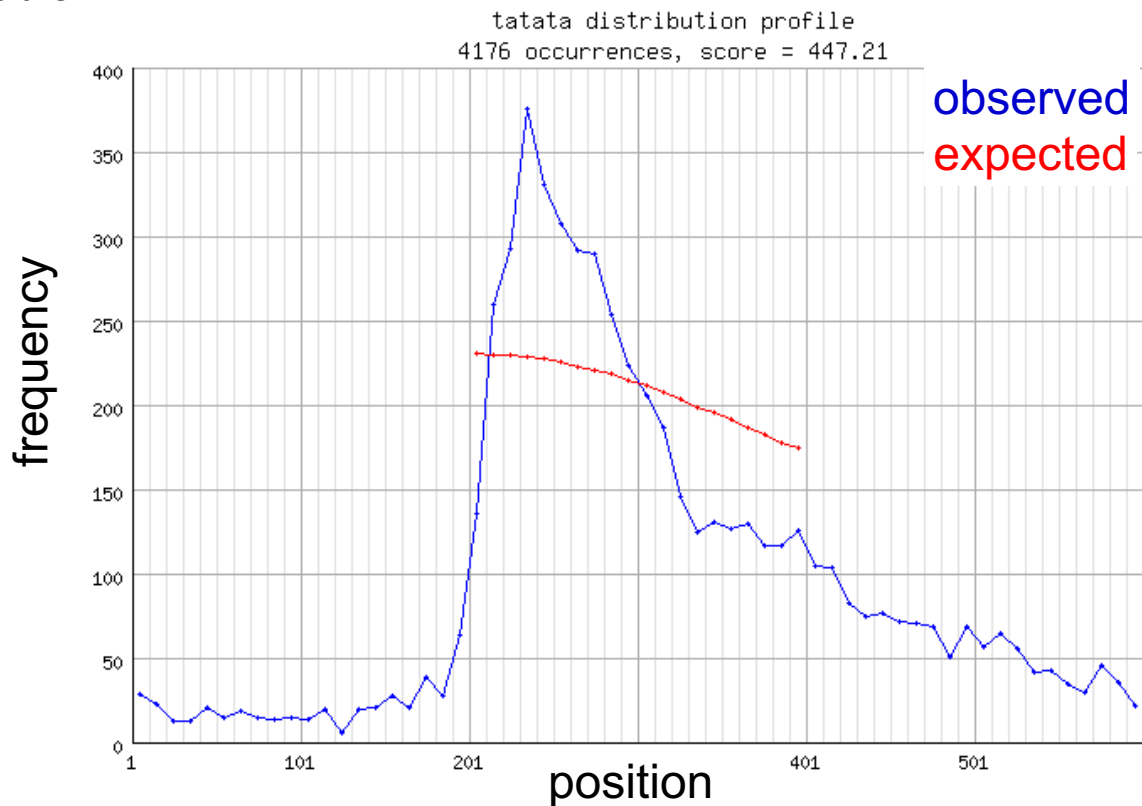
String-based motif discovery: weaknesses

- No direct treatment of motif degeneracy
 - NB: degenerated words can be analyzed with similar statistics, but it is not tractable due to the increase of the number of motifs: 15k possible words of length k.
- String motifs are poor descriptions for genome-scale motif matching.
 - Matrices are more appropriate to describe the weight of each substitution at a given position.
- Solution
 - string-based approach for motif discovery (RSAT programs *oligo-analysis*, *dyad-analysis*, *position-analysis*, *local-words*).
 - use discovered strings as seeds for building a matrix, which can be used for motif search (RSAT program *matrix-from-motifs*)

Position-analysis

Word position distribution

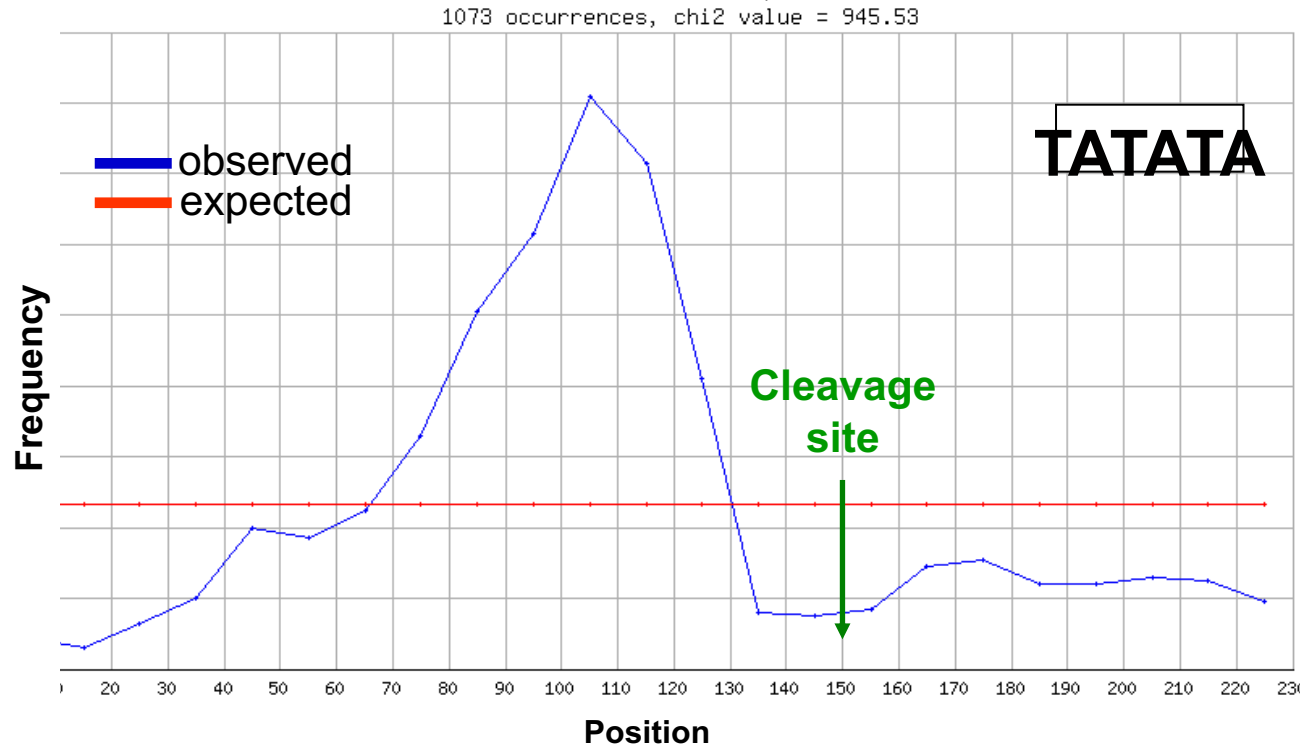
- Positions relative to the stop codon



- van Helden, J., del Olmo, M. and Pérez-Ortín, J.E. (2000) Statistical analysis of yeast genomic downstream sequences reveals putative polyadenylation signals. *Nucleic Acids Res*, **28**, 1000–1010.

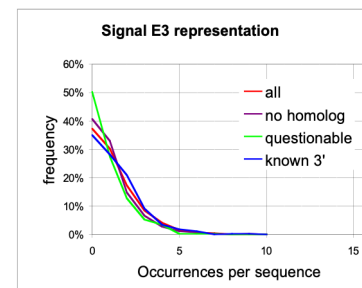
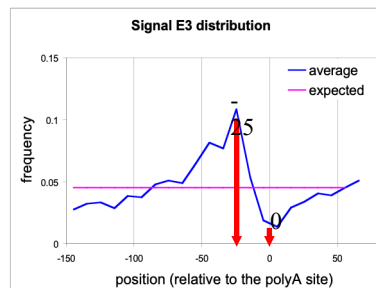
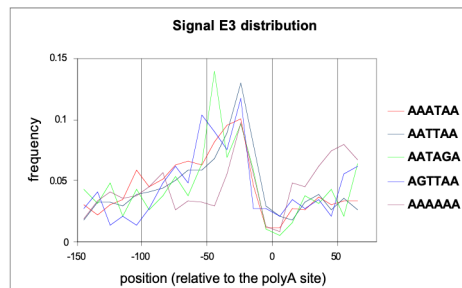
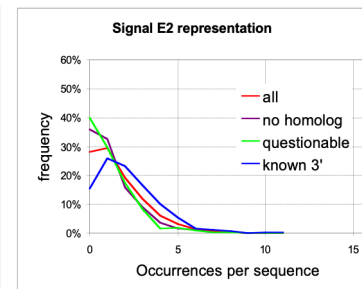
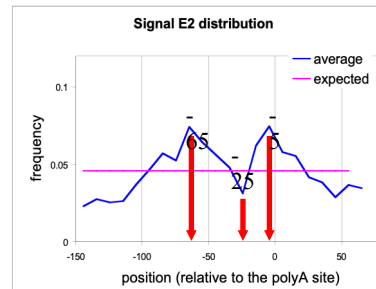
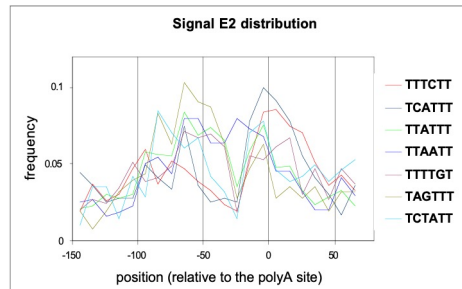
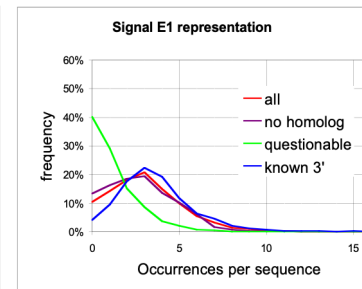
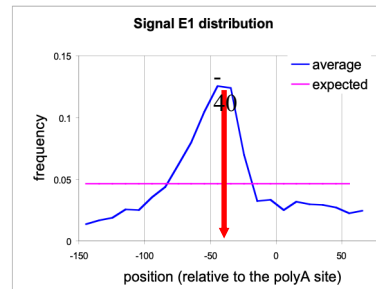
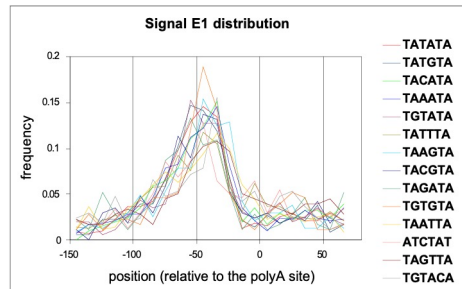
Profiles of hexanucleotides distribution around 1500 yeast TSS

- Positions relative to the cleavage site



- van Helden, J., del Olmo, M. and Pérez-Ortín, J.E. (2000) Statistical analysis of yeast genomic downstream sequences reveals putative polyadenylation signals. *Nucleic Acids Res*, **28**, 1000–1010.

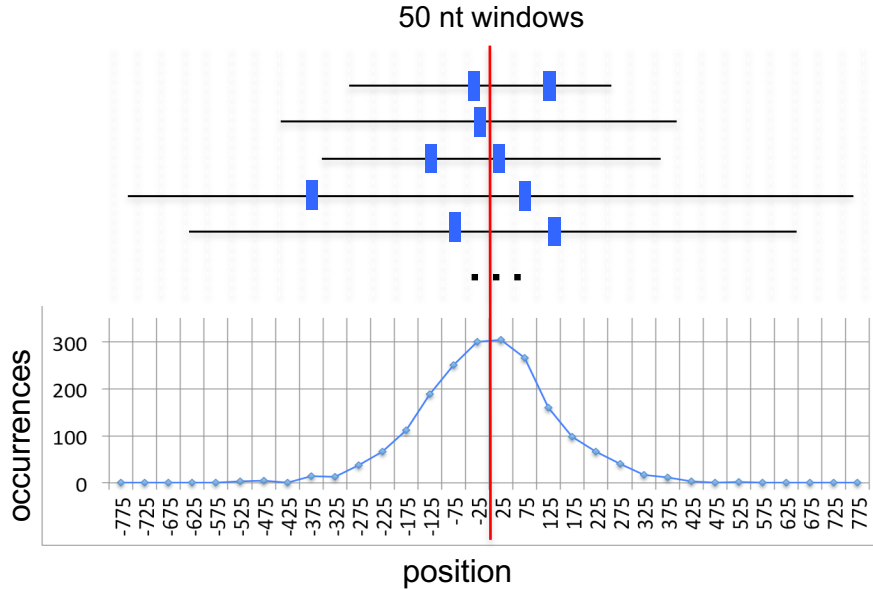
Clusters of positionally biased k -mers around termination sites



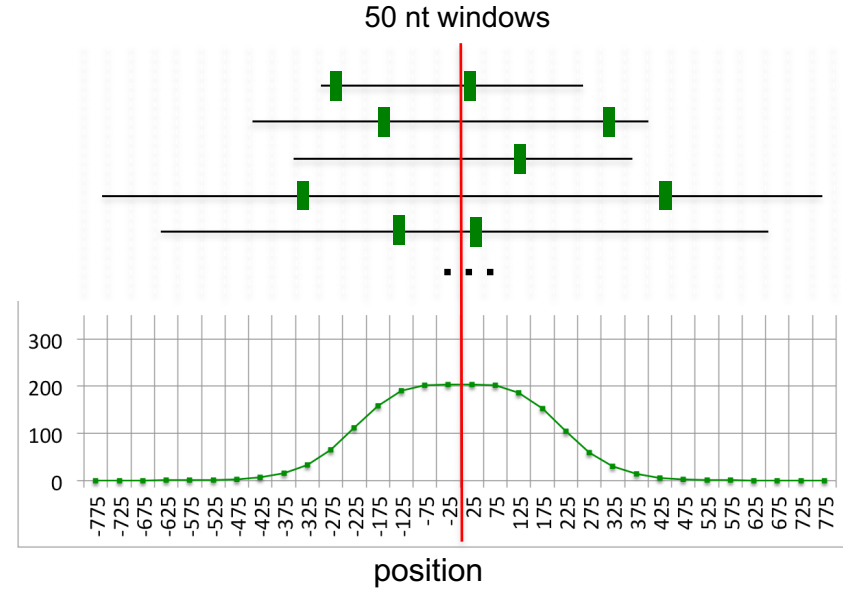
Detecting heterogeneous repartition along sequences

■ ■ 7-mer (e.g.AACAAAG)

Observed occurrences per window



Expected occurrences per window according to a homogeneous model



Drawing by Elodie Darbo

position-analysis method

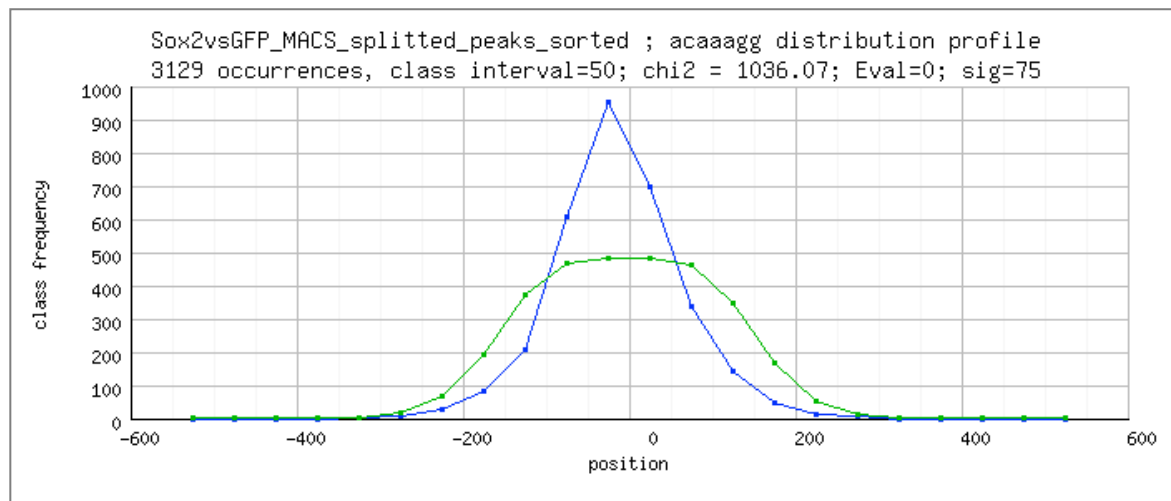
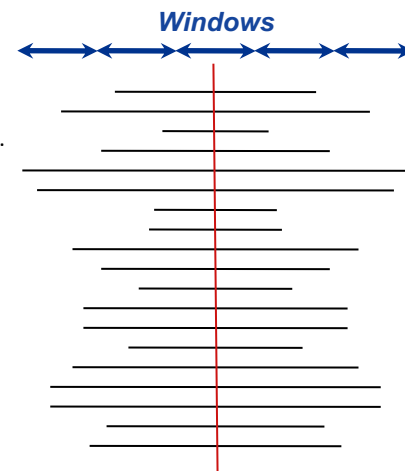
- van Helden, J., del Olmo, M. and Perez-Ortin, J. E. (2000). Statistical analysis of yeast genomic downstream sequences reveals putative polyadenylation signals. *Nucleic Acids Res* 28, 1000-10.

Application to chip-seq:

- Thomas-Chollier M, Herrmann C, Defrance M, Sand O, Thieffry D, van Helden J. (2012). RSAT peak-motifs: motif analysis in full-size ChIP-seq datasets. *Nucleic Acids Res* 40(4): e31.
- Thomas-Chollier M, Darbo E, Herrmann C, Defrance M, Thieffry D, van Helden J. (2012). A complete workflow for the analysis of full-size ChIP-seq (and similar) data sets using peak-motifs. *Nat Protoc* 7(8): 1551-1568.

Detecting biases in word positions

- The program position-analysis (van Helden et al., 2000) detects words showing a heterogeneous distribution of occurrences across a set of input sequences.
- Principle: for each word
 - Compute the number of occurrences in non-overlapping windows starting from a reference point (sequence start, center or end).
 - Compute the expected occurrences in each window according to a homogeneous distribution model.
 - Compute the difference between the observed and expected positional distribution (chi2 test for goodness of fit).
- Example: Sox2 peaks from Chen, 2008
 - 10,929 peaks of size between 60 and 1,059 bp
 - Word length $k=7$
 - Reference position: the center of each peak.
 - The most significant word is ACAAAGG, which corresponds to the Sox2 consensus.



- **Green: expected occurrences**
 - Note: the expectation decreases with the distance to peak center because peaks have variable lengths.
- **Blue: observed occurrences**
 - The word ACAAAGG is concentrated the center the ChIP-seq peak regions.

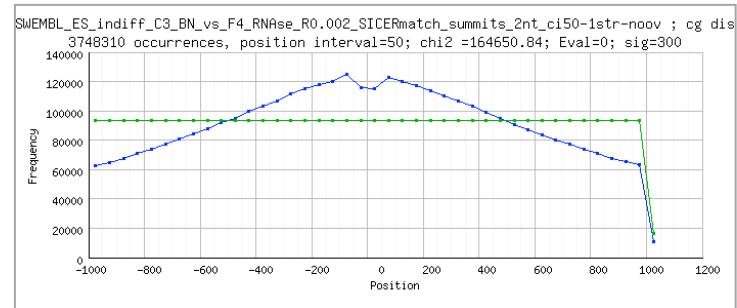
Position-analysis of dinucleotides around replication origins

- 65,009 peaks
- 2kb on each side of peak summits (130Mb analyzed in total)
- K-mer occurrences per 50bp windows
- **Background model: homogeneous distribution**
- Significance computed with Chi-square conformity test.
- Result: **all** dinucleotides are completely biased, with p-values < 1e-300.

Sequence	ID	Occ	Overlaps	Chi2	df	Pval	Eval	Sig	Rank
cg	cg	3748310	0	164650.8	40	0.0e+00	0	300.0	1
cc	cc	8078476	2609220	78471.5	40	0.0e+00	0	300.0	2
gg	gg	8056188	2595227	77779.8	40	0.0e+00	0	300.0	3
ta	ta	5242474	0	72304.9	40	0.0e+00	0	300.0	4
aa	aa	6153023	2033169	66112.4	40	0.0e+00	0	300.0	5
tt	tt	6178248	2048441	65633.9	40	0.0e+00	0	300.0	6
gc	gc	8512412	0	64740.0	40	0.0e+00	0	300.0	7
at	at	6039429	0	58647.8	40	0.0e+00	0	300.0	8
tc	tc	8137303	0	26051.4	40	0.0e+00	0	300.0	9
ga	ga	8101277	0	25343.6	40	0.0e+00	0	300.0	10
ag	ag	10092541	0	21823.5	40	0.0e+00	0	300.0	11
ct	ct	10113605	0	21797.1	40	0.0e+00	0	300.0	12
ac	ac	6833408	0	15129.5	40	0.0e+00	0	300.0	13
gt	gt	6841055	0	14892.4	40	0.0e+00	0	300.0	14
ca	ca	9621040	0	13119.9	40	0.0e+00	0	300.0	15
tg	tg	9613852	0	12918.0	40	0.0e+00	0	300.0	16

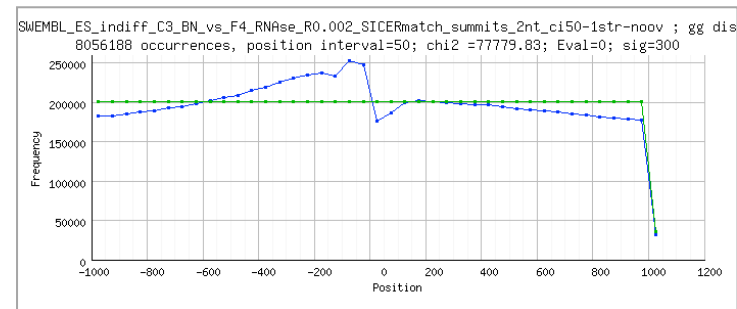
CG

- observed
- expected



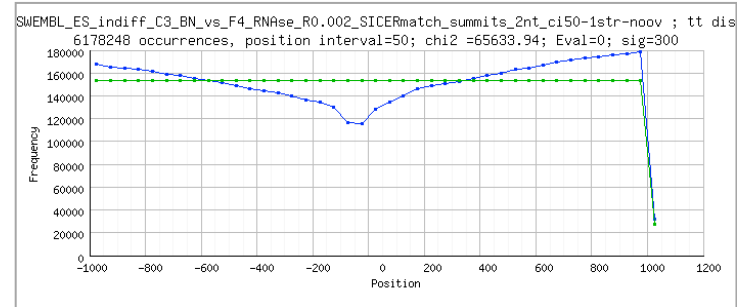
GG

- observed
- expected



TT

- observed
- expected



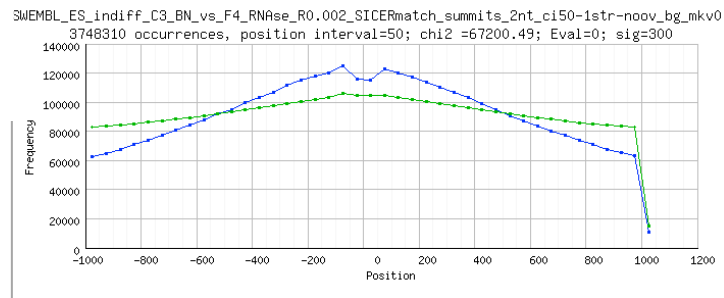
Position-analysis of dinucleotides around replication origins

- 65,009 peaks
- 2kb on each side of peak summits (130Mb analyzed in total)
- K-mer occurrences per 50bp windows
- **Background model: window-specific estimation based on nucleotide composition.**
- Significance computed with Chi-square conformity test.
- Result: **all** dinucleotides are completely biased, with p-values < 1e-300.

Sequence	ID	Occ	Overlaps	Chi2	df	Pval	Eval	Sig	Rank
cg	cg	3748310	0	67200.5	40	0.0e+00	0	300.0	1
ac	ac	6833408	0	10173.3	40	0.0e+00	0	300.0	2
gt	gt	6841055	0	10001.6	40	0.0e+00	0	300.0	3
ca	ca	9621040	0	8646.5	40	0.0e+00	0	300.0	4
tg	tg	9613852	0	8508.0	40	0.0e+00	0	300.0	5
ta	ta	5242474	0	7453.5	40	0.0e+00	0	300.0	6
tt	tt	6178248	2048441	4902.7	40	0.0e+00	0	300.0	7
aa	aa	6153023	2033169	4628.1	40	0.0e+00	0	300.0	8
gg	gg	8056188	2595227	3843.9	40	0.0e+00	0	300.0	9
cc	cc	8078476	2609220	3773.6	40	0.0e+00	0	300.0	10
at	at	6039429	0	1763.3	40	0.0e+00	0	300.0	11
gc	gc	8512412	0	1447.1	40	0.0e+00	0	300.0	12
ag	ag	10092541	0	1392.3	40	0.0e+00	0	300.0	13
ct	ct	10113605	0	1367.4	40	0.0e+00	0	300.0	14
tc	tc	8137303	0	1027.4	40	0.0e+00	0	300.0	15
ga	ga	8101277	0	887.6	40	0.0e+00	0	300.0	16

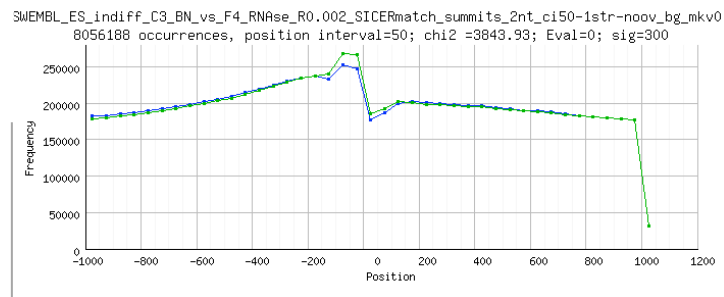
CG

- observed
- expected



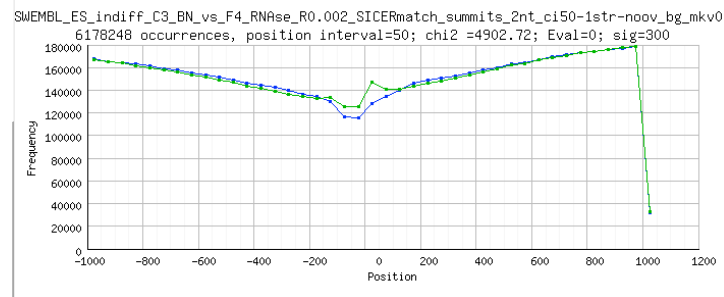
GG

- observed
- expected



TT

- observed
- expected

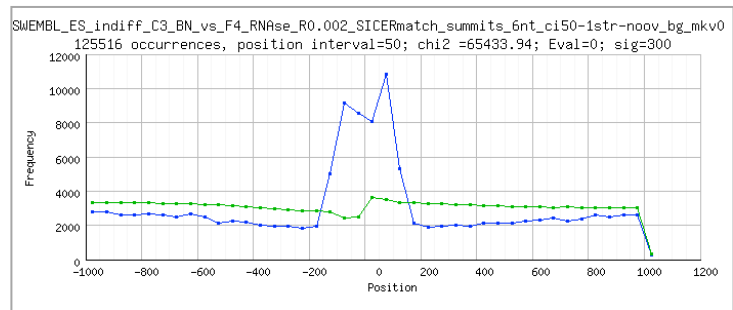


Position-analysis of hexanucleotides around replication origins

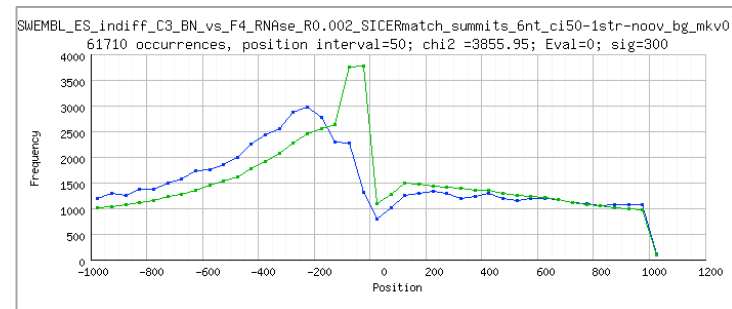
- **Background model: window-specific estimation based on nucleotide composition.**
- A lot of very highly significant 6-mers.
- Most of them are low-complexity motifs (periodic k-mers).

Sequence	ID	Occ	Overlaps	Chi2	df	Pval	Eval	Sig	Ran
acacac	acacac	125516	128532	65433.9	40	0.0e+00	0	300.0	1
gtgtgt	gtgtgt	125740	127945	62813.5	40	0.0e+00	0	300.0	2
cacaca	cacaca	143816	131933	57978.6	40	0.0e+00	0	300.0	3
tgtgtg	tgtgtg	143246	130948	56183.5	40	0.0e+00	0	300.0	4
gggggg	gggggg	61710	64984	3855.9	40	0.0e+00	0	300.0	5
cccccc	cccccc	61215	65010	3540.8	40	0.0e+00	0	300.0	6
tttttt	tttttt	73687	122406	2182.5	40	0.0e+00	0	300.0	7
aaaaaa	aaaaaa	73293	122290	2055.5	40	0.0e+00	0	300.0	8
gggagg	gggagg	141834	13752	1957.4	40	0.0e+00	0	300.0	9
tggggg	tggggg	100645	0	1936.2	40	0.0e+00	0	300.0	10
ggggga	ggggga	79273	0	1932.2	40	0.0e+00	0	300.0	11
tccccc	tccccc	80747	0	1923.7	40	0.0e+00	0	300.0	12
ggaggg	ggaggg	127623	13258	1919.7	40	0.0e+00	0	300.0	13
ccccca	ccccca	101109	0	1872.0	40	0.0e+00	0	300.0	14
tgtgta	tgtgta	49360	0	1850.7	40	0.0e+00	0	300.0	15
cctccc	cctccc	142300	13511	1823.3	40	0.0e+00	0	300.0	16
atgtgt	atgtgt	51630	0	1812.9	40	0.0e+00	0	300.0	17
gggtgg	gggtgg	104120	6751	1799.7	40	0.0e+00	0	300.0	18
acacat	acacat	51430	0	1799.4	40	0.0e+00	0	300.0	19
tacaca	tacaca	48626	0	1782.6	40	0.0e+00	0	300.0	20
ccctcc	ccctcc	128878	12851	1755.2	40	0.0e+00	0	300.0	21
ccaggg	ccaggg	93739	0	1706.5	40	0.0e+00	0	300.0	22
cgccgc	cgccgc	52126	8980	1704.4	40	0.0e+00	0	300.0	23
ctcccc	ctcccc	111276	4097	1658.5	40	0.0e+00	0	300.0	24
ccctgg	ccctgg	93919	0	1635.7	40	0.0e+00	0	300.0	25

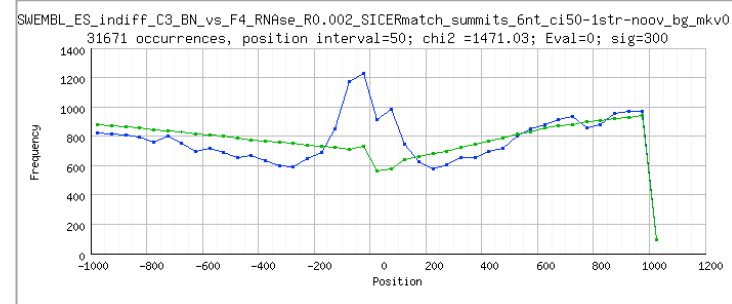
ACACAC
- observed
- expected



GGGGGG
- observed
- expected



GTGTAT
- observed
- expected



Examples of applications

Analysis of cell cycle data : results

family	oligo-analysis				dyad-analysis (non-coding dyad frequency calibration)			
	word	reverse clpt	sig	remark	dyad	reverse clpt	sig	remark
CLN2	TACGCGAA . . TTCGCGTA		30.5	MBF; SBF variant	TTTACGCGAAAA TTTTCGCGTAAA		29.0	MBF; SBF variant
	TACGCGTA . . TACGCGTA		30.5	MBF; SBF	GAAAACGCGTAAA TTTACGCGTTTTC		29.0	MBF; SBF
	TTTCGCGTCG CGACGCGAA		30.5	MBF; SBF variant	TTTTTCGCGTCA . . TGACGCGAAAA		29.0	MBF; SBF variant
	AAACGCGAA . . TTCGCGTTT		30.5	MBF; SBF variant	TTTACGCGTCA . . TGACGCGTAAA		29.0	MBF; SBF
	TTTCGCGTCA . TGACGCGAA		30.5	MBF; SBF variant	CGACGCGAAAA TTTTCGCGTCG		29.0	MBF; SBF variant
	TGCCAA TTGGCA		1.8		GAAAACGCGTCA . . TGACGCGTTTTC		8.1	MBF; SBF
	ATCAAG CTTGAT		1.3		AAAn8CGC GCGn8TTT		1.9	
Y'	CTCGTC GACGAG		1.8		CAAn5CGC GCGn5TTG		1.1	
	AGTATC GATACT		1.2		AGTnGAG CTCnACT		3.0	
					CAGn{10}ATC GATn{10}CTG		2.0	
histone	CGCCCG CGGGCG		2.6		ATCn{12}GAG CTCn{12}GAT		1.2	
	CCAGAA TTCTGG		1.7	Mcm1	GCGn8AGAAC GTTCTn8CGC		3.0	
					CGCCCG CGGGCG		1.3	
Cell cycle MET	TGCCACAGTT AACTGTGGCA		10.1	Met31; Met32	ATTn2GCG CGCn2AAT		1.3	
	TCACGTGA TCACGTGA		10.1	Met4/Met28/Cbf1	GCCACAGTT AACTGTGGC		8.6	Met31; Met32
	ACAGAG CTCTGT		1.9		GTCACGTGAC GTCACGTGAC		6.9	Met4/Met28/Cbf1
	GACTCA TGAGTC		0.9					
CLB2	CCAAAG CTTTGG		1.3		CCCN6GAA TTCn6GGG		2.5	ECB
	CCTTCA TGAAGG		0.9	NEG	CAAn13GCC GGCn13TTG		0.9	
MCM	AGAGCA TGCTCT		1.4		ACCn14AAT ATTN14GGT		0.9	
	TCCTAA TTAGGA		1.0	Mcm1	TCCCN4GGA TCCCN4GGA		3.9	ECB variant
SIC1	AACCAGCAA TTGCTGGTT		20.0	Swi5; Ace2	AAAnAGG CCTnTTT		2.8	ECB ?
	AGCCAGCAA TTGCTGGCT		20.0	Swi5; Ace2	AGGn10ACT AGTn10CCT		1.2	
	AACCAGCC GGCTGGTT		8.0	Swi5; Ace2	AACCAGCA TGCTGGTT		20.0	Swi5; Ace2
					AACCAGCCAGCA TGCTGGCTGGTT		20.0	Swi5; Ace2

■ Gene clusters from Spellman et al. (1998). Mol Biol Cell 9(12), 3273-97

Plasmodium erythrocytic cycle

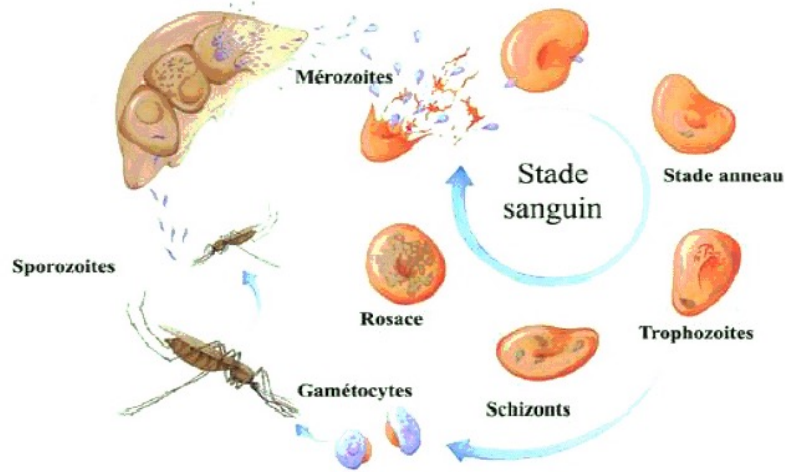
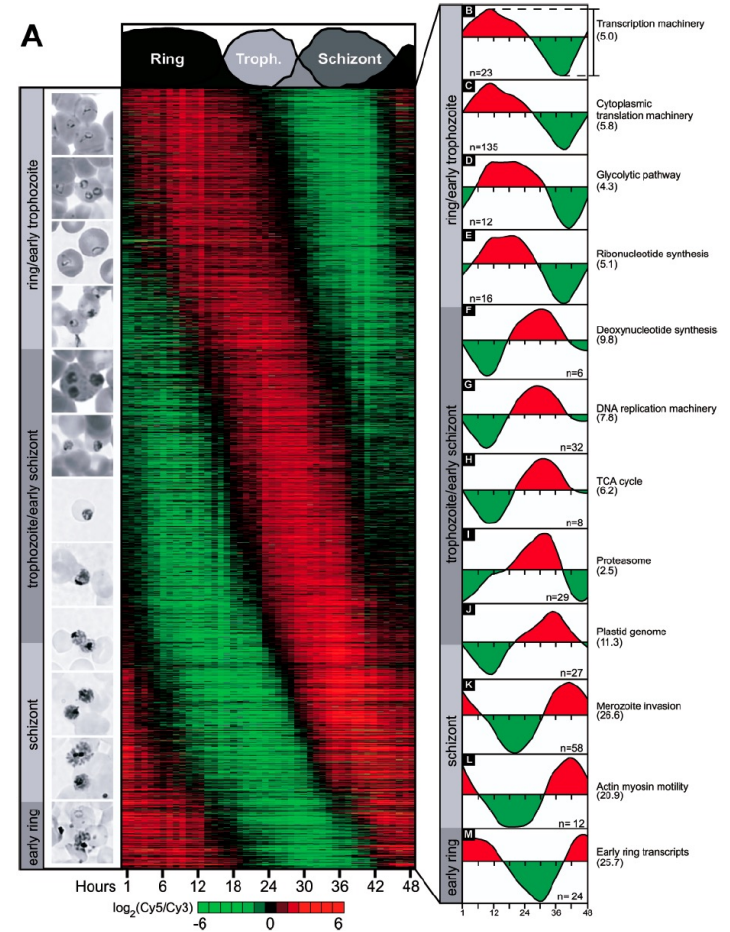
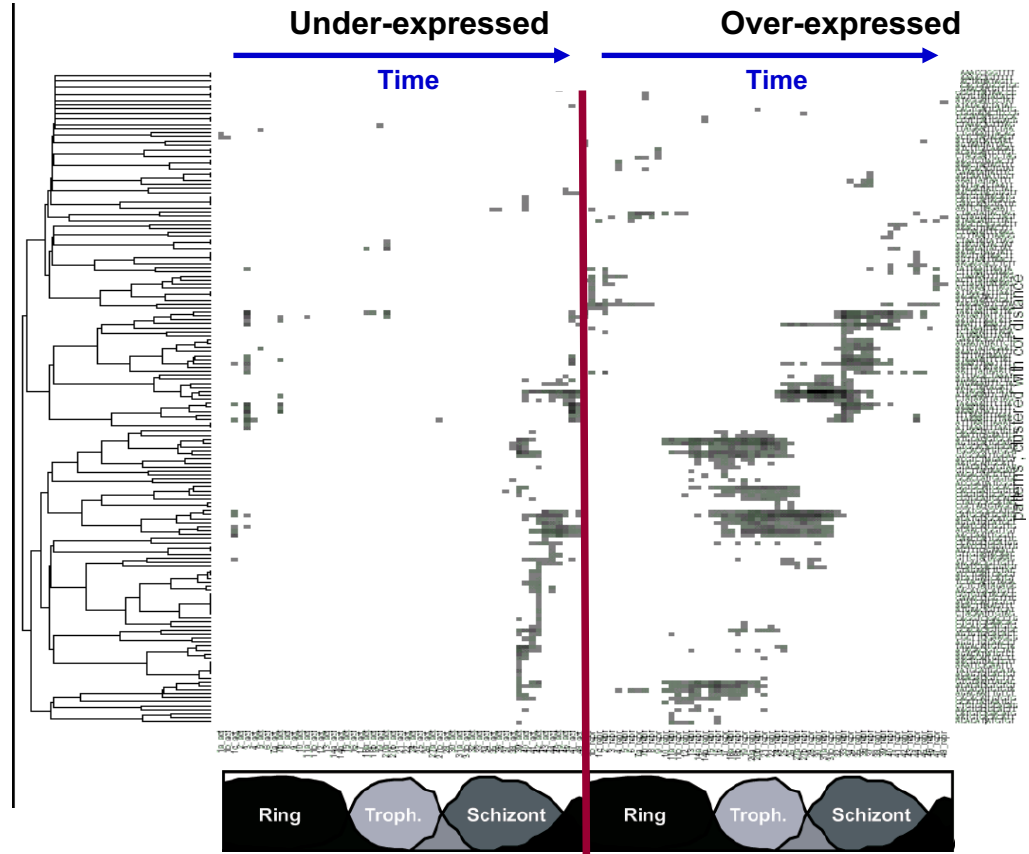


figure 2: Cycle de vie de *Plasmodium falciparum* (source :institut pasteur (France) page web)



Over-represented oligos in promoters of under- and over-expressed genes at different time points of the erythrocytic cycle of *Plasmodium falciparum*



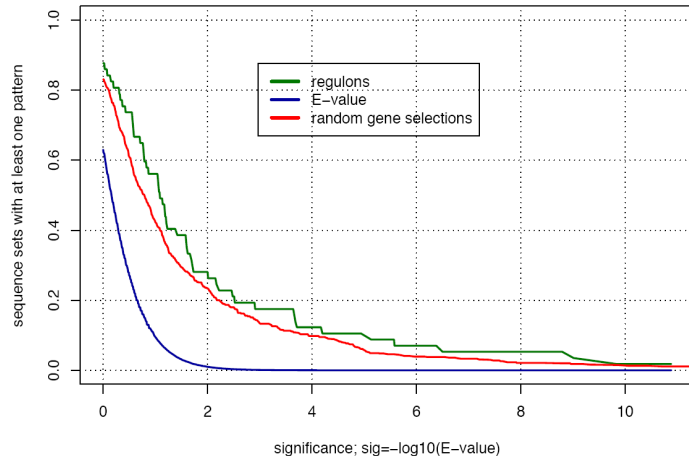
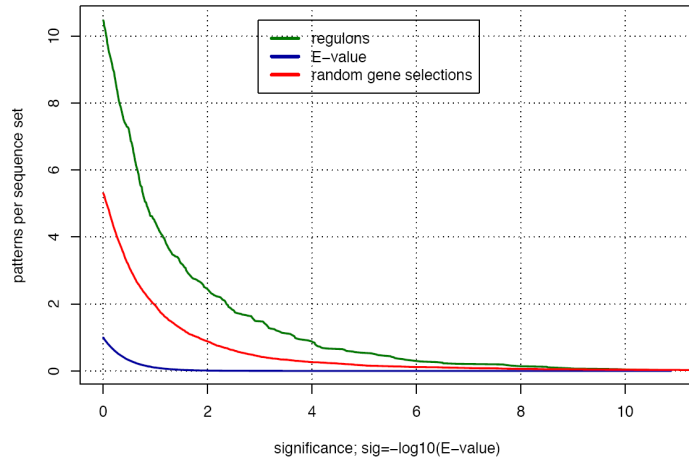
Supplementary material

Motif discovery: string-based algorithms

- Count occurrences observed for each word
- Calculate expected word frequencies
 - Choice of a model :
 - independently distributed nucleotides (equiprobable or biased alphabet utilization)
 - Markov chain : on basis of subword frequencies
 - External reference (e.g. word frequencies observed in the whole set of upstream sequences)
- Calculate a score for each word
 - obs/exp ratio (very bad)
 - log-likelihood
 - Z-value
 - binomial probability
- Select all words above a defined threshold
 - Statistical criterion for establishing the threshold

Motif significance in regulons - *Homo sapiens*

Homo_sapiens_Ensembl ; oligos ; 50 regulons ; 507 random selections



■ In *Homo sapiens*

- The rate of false positive is much higher than the theoretical expectation
- The number of motifs detected in regulons is still higher, but the significance score is quite inefficient to distinguish between reliable motifs and false positives.
- This indicates that the background model is inadequate to treat the complexity of human promoters.